The Λ-Anchored Acceleration Scale $a_0 = c^2\sqrt{\Lambda/32\pi}$: A Completed Falsification Map, Exact No-Go Results, and the Specification of the Unique Surviving Object

Carl P. Zimmerman

June 2026

June 2026, second edition, final pass. Whitepaper prepared for Zenodo deposit. Repository: `zimmerman-formula` *(commit-hash provenance in Appendix A).*

Abstract

We report the completed falsification map of a research program organized around a single phenomenological kernel: the identification of the MOND acceleration scale with the cosmological constant through $a_0 = c^2\sqrt{\Lambda/32\pi} = (c/2)\sqrt{G \cdot \rho_{\mathrm{DE}}} = cH_\Lambda/Z$, with $Z = \sqrt{32\pi/3} = 5.789$ and $a_0 = 9.36\times10^{-11}\ \mathrm{m\,s^{-2}}$. The kernel's empirical standing is strong and convention-robust: it fits the SPARC radial-acceleration relation at 0.105 dex through a de Sitter–Unruh modified-inertia construction, and under the standard unweighted dex-scatter metric the framework value sits within 0.3% of the SPARC-optimal scatter. What this paper establishes is threefold. First, a closed perimeter over published covariant hosts: every published covariant realization of MOND-scale phenomenology now fails a named, quantified gate — AeST at the Cassini external-field quadrupole; single-scalar Galileon/k-mouflage hosts at the Bruneton–Esposito-Farèse singular surface together with the GW170817 and ISW gates; and the Deffayet–Woodard nonlocal-metric class at a Cassini quadrupole computed here for the first time ($Q_2 = +1.74$ to $+2.80\times10^{-26}\ \mathrm{s^{-2}}$, 8.8–14.6$\sigma$ against the 2026 bound), with the surviving function-family residue explicitly characterized as a g_{ext}-fragile, RAR-penalized, theory-unselected tuning sliver. Second, a set of exact results on the only remaining class, modified inertia: the Milgrom (1999) vacuum-response ansatz is excluded by planetary ephemerides by a factor ∼54,000; a mutual-exclusion no-go closes its natural coupling family; the data-selected susceptibility coupling F4 survives five kill-tests but is then excluded as a solar-system worldline law by a nonlinear ephemeris-fit emulation of the Sun's anomalous reflex at both candidate normalizations; its proposed bath mechanism is refuted at all orders in the coupling by a trajectory-blindness lemma; and the unique field-side escape (non-Huygens fields) is mapped exactly — a closed-form commutator showing trajectory-blindness genuinely breaks, a memory Langevin theory with a universal p = 2 frequency gate and a non-empty knee window, a MOND-signed inertia deficit (the program's only right-signed bath channel), and four independent magnitude walls (margins 10^5–10^{86}) that close every bath-coupled point-worldline realization. Third, a data program: a wide-binary methodology result establishing that the current Gaia DR3 observable is degeneracy-limited, and an independent re-measurement and hostile audit of the weak-lensing RAR early/late split showing the strongest standing exposure against the framework *hardens* under its most plausible systematics (8.6–9.2σ on the u−r axis), together with a 40.5σ exclusion of metric-passive (baryon-only-lensing) modified inertia on the same data. The residue of the map is a sharply specified missing object: a field-level hybrid with a trajectory-nonlocal, acceleration-keyed, MOND-signed matter sector dressed by at most one power of frequency, an ultralight knee scale $mc^2 \in [1.3\times10^{-29},\ 1.6\times10^{-24}]$ eV, a $\kappa = \sqrt{a^2 + (cH)^2}$ kernel that supplies the correct length scale, and a separate lensing-carrying partner. The specification is then itself confronted, and the confrontation is part of the result. A gated candidate matrix over the published field-level literature returns an empty field: best live score 3/8; the only 5/8 scorer (AeST) is dead on the spec's own named

killer gate, so the matrix is gated, not gerrymandered; and the spec's two *derived* ingredients — the inertia-side lever and the ultralight knee field — appear in no published theory, live or dead. The banked kill-tests, run numerically on the nearest published cousin (Berezhiani–Khoury superfluid dark matter), return a complementary failure: the class clears exactly the two lensing walls that kill universal-force MOND — including the type split, sign-matched with the measured +0.261 dex inside the predicted bracket, though the mechanism's sharp condensation-staircase form is refuted at $\approx 7.3\sigma$ on the mass-binned profiles — and dies face-value in the solar system where modified inertia survives (Cassini quadrupole 24–39σ, via a new exact external-field-independent $q_{\rm ph} = -3/7$), escaping only by incompleteness. And the first construction attempt at the spec itself died pre-assembly: the fraction-limited ultralight carrier cannot supply the MOND amplitude — the sourcing route needs carrier fractions f $\approx$ 0.85–3.3 (clustered) to $\sim 10^2$–10^4 (homogeneous) against a $\lesssim$1–3% structure-formation ledger; the mediator route falls 4.2 dex short at maximum charity and 6.7–7.0 dex with Cassini enforced; and the MOND sign and the knee scale cannot come from the same field's vacuum tail. We state with equal weight what is *not* established: no derivation of the coefficient Z = 5.789 exists; no mechanism for the kernel exists (every candidate computed to date is refuted or magnitude-blocked); the a_0(z) evolution claim is a conditional ansatz, not a derivation; and the field-level hybrid has no surviving realization — its first construction died pre-assembly, and what remains of the specification is a carrier whose amplitude must be paid by a structure no published theory supplies. This paper claims a map, exact kills, and a specification — not a theory of everything.

Second edition. Nine further verdicts were banked after the first edition closed; this edition adds one section (Section 7) consolidating them into the program's spine: a derivation chain from Λ to the named effective law, continuous end to end and broken in exactly three places — the coefficient, the mechanism, and the quantum gate — each break now bounded by a theorem or a live contest rather than by ignorance. Three new exact results anchor it. The Deser–Levin temperature is identically the Tolman-shifted Gibbons–Hawking temperature ($\sqrt{a^2+H^2}\cdot|\xi| = H$ exactly), so Jacobson's Clausius construction run with the de Sitter worldline temperature reproduces Einstein unchanged at all orders: horizon thermodynamics provably cannot reach MOND — Clausius consumes T where MOND needs dT/da. A double-counting theorem excludes real-mass lensing partners as a class for modified-inertia dynamics (monotone μ forces the partner density to zero wherever kinematics probes, against the 40.5σ lensing demand; 8.7–21.6σ on our own SPARC pipeline under every convention). And the worldline-bath closure now spans point, composite, and extended detectors, the last through the exact identity $\varepsilon = GMH/c^3 = r_g/2R_H$: the coherent N^2 gain and the WEP violation are the same number. The coefficient contest is left honest: Z = 5.789 data-certain but underived; Verlinde's conditional 6 (dimension-forced given five named postulates) the only derivation-flavored candidate; the two empirically degenerate at 3.65%; a pre-registered geometry audit null — nothing forces either. The missing object is now named in both halves — a khronon-framed covariantization of Milgrom's 2022 time-nonlocal modified inertia with the exponential μ-tail as the matter sector (buildable on paper, unbuilt, gates named), and a metric-level Ψ-channel lens-only slip as the partner (the unique surviving class by theorem, unbuilt, gates named) — a blueprint, specified, not found. And the quantum gate moved by the program's own pre-registered discriminator: the DSSYK vacuum-placement contest remains terminal at the algebra level, but the edge placement fails the de Sitter quasinormal ladder structurally under both dimensional matchings, so the contest collapses toward the MOND-favorable center for the sign-relevant observable. The boundary is unchanged: no derivation of Z exists; no mechanism survives at the worldline; the hybrid is unbuilt. This edition adds a chain and a blueprint, not a theory.

Final pass. Five further verdicts complete the campaign. The matter half of the blueprint is now **built at the equation-of-motion level**: the Galley/Schwinger–Keldysh doubling of Milgrom-2022 yields a strictly retarded, u-clocked causal equation of motion — banked solar numbers reproduced to 0.02–0.03%, the energy ledger closed onto the khronon at 10^{-14}, zero pre-acceleration — together with the boundary theorem the field had been missing (Theorem X2: causality plus vacuum passivity force $\hat{\mu}(0) \geq \hat{\mu}(\infty)$, while the deep-MOND limit of *any* modified-inertia theory forces the inversion — no passive vacuum closes any causal MI; the channel is irreducibly active, and the Λ/dS bath is the only reservoir in the action that can pay, with $\times 10^2$–10^4 to spare). The mechanism break thereby collapses to one object with one property, at theorem-plus-data grade: an active, dS-invariance-breaking medium — the khronon sector itself, Λ-pumped — whose required spectral density $\sigma_{\rm req}$ is computed and stands as the sole derivation target; the sign is two-method confirmed (the inertia-deficit channel is $M^2 < 2H^2$ — intrinsically de Sitter), and the last legal escape (a dS-invariant positive spectral mixture) is data-closed on both pre-registered arms at once (+0.023–0.026 dex above the locked SPARC baseline — past the pre-registered closure line — *and* ×21 over the solar-reflex budget, through a forced-positive variance pincer with no sweet spot). The lensing carrier narrows constructively: the scalar realization of the unique slip class is closed by four machine-derived walls — not ghosts — leaving vector/spin-2-on-u or nonlocal carriers, with $c_T \equiv 1$ and $\alpha_M \equiv 0$ inherited identically and the type-split dial found in the surviving generators themselves (field-line-bending operators: right sign and range for the measured type-irreducible +0.194 dex). The two halves are assembled into one four-sector action — a candidate-theory scaffold, named, specified, gated — not validated — with exactly one underived number. And the mechanism class yields a new falsifiable prediction: no kernel produces the $\sqrt{a/a_0}$ onset, so μ must flatten to constant + $O(a^2)$ below some $a_\star$. The boundary stands: Z underived; $\sigma_{\rm req}$ underived; the quantum gate unmoved at the algebra. **Addendum (same day).** A hostile audit independently re-derives Theorem X2 by a bath-microphysics route and *extends it to nonlinear passive media* (energy conservation alone closes the loophole), and finds the construction's signed energy ledger draining the reservoir secularly in exactly the direction the theorem demands; the new flattening prediction survives its first confrontation with existing deep kinematics untested in its allowed window ($a_\star \leq 0.107\, a_0$ at 95%); and the published JWST-era high-z kinematics are shown to sit in a regime ($g_{\rm obs} = 5.7$–$24\, a_0$) where no $a_0(z)$ branch can yet be discriminated — with the decisive future measurements named and registered in both cases. **Second addendum (the construction round, same day).** Five further machine verdicts land. The mechanism slot becomes a **defined calculation**: the khronon evades the positivity no-go outright (no Källén–Lehmann representation exists for it; the replacement two-variable representation is derived; the broken symmetry's dilatation orbits are shown to *be* the Deser–Levin trajectory family, exactly), the free khronon provably cannot carry the kernel (the pump must modify the dynamics — the activity theorem's third independent derivation), and the required pump fingerprint is computed with acceptance conditions written. The carrier search hardens into two theorems — the keying theorem (any *local* acceleration-keyed slip carrier pollutes the matter channel, scalar and vector alike) and static equivalence (the entire time-nonlocal history-keyed class inherits the same walls verbatim; a mid-day convergence conjecture is computed false and retired same-day) — leaving the lensing exposure unexplained by every construction route explored, with the survivor space named. The slip sector's naive cosmological extension is killed convention-robustly (E_G off by 73σ; the only viable global cap would freeze the entire galactic RAR), banking a second-discriminant requirement, currently unowned. And the causal construction's transient fingerprint is found operator-form-dependent (the action EOM gives suppression where the implicit-μ reading gave the banked ×2.32) and is confronted with tidal-dwarf data for the first time: the naive transplant dies at $\sim 11\sigma$; the construction's own EFE-embedded prediction passes; the decisive three-way test is registered.

Table of Contents

1. Introduction and scope
2. The kernel and its empirical standing
3. The covariant-host trilemma: every published host fails a named gate
4. The modified-inertia program: selection, kills, and exact no-go results
5. The data program: wide binaries and the lensing RAR
6. The specification of the missing object
7. The derivation chain *(second edition; final-pass forms in 7.3–7.6)*
8. Falsifiable predictions and decision dates
9. Methods appendix: the discipline
Appendix A: commit-hash provenance

References

1 Introduction and scope

The acceleration scale $a_0 \approx 1.2\times10^{-10}$ m s^{-2} that organizes galaxy dynamics (Milgrom 1983, ApJ 270, 365) is numerically close to acceleration scales built from the cosmological constant. This program takes one specific identification seriously as a falsifiable kernel,

$$a_0 = c^2\sqrt{\Lambda/32\pi} = (c/2)\sqrt{G\cdot\rho_{\rm DE}} = cH_\Lambda/Z, \quad Z = \sqrt{32\pi/3} = 5.789, \quad a_0 = 9.36\times10^{-11}\ \mathrm{m\,s^{-2}},$$

footed on the dark-energy density alone ($\rho_{\rm DE} = \Omega_\Lambda\rho_{\rm crit}$, the pure-$\Lambda$ rate H_Λ), not on $\rho_{\rm total}$, and subjects it — and every covariant or worldline-level theory that could host it — to a uniform adversarial protocol (Section 9). The program's products are of three kinds: confrontations of published hosts with decisive data gates; exact mathematical results (closed-form kernels, no-go lemmas) for the mechanism class the kernel itself suggests; and pre-registered confrontations of the kernel with public data.

The boundary of this paper, stated before any result. Nothing below claims a complete theory. The coefficient Z = 5.789 is data-selected; a multi-route audit found no framework that forces it, and none of the dozens of raw coefficients produced by the program's exact calculations lands on Z or any function of it (each such non-coincidence is recorded at the point it arises, so that no later reader "discovers" one). No mechanism for the kernel survives the program's own kill-tests. The distinctive a_0(z) evolution claim is held as a conditional ansatz with its conditionality string attached (Section 2.3). What the program *has* produced is, we believe, of independent value regardless of the kernel's eventual fate: a completed falsification perimeter over the published covariant hosts, several exact results that constrain any future modified-inertia construction, two data-methodology results (wide-binary degeneracy; lensing-RAR conversion audit) addressed to both sides of the respective disputes, and the most precise specification yet stated of the object that would have to exist for this class of phenomenology to have a dynamical basis.

A reader who wants the single-sentence verdict: the perimeter closed, the kills are clean, the kernel survives as phenomenology, and exactly one object — specified in Section 6 — remains; its first construction attempt died pre-assembly (Section 6.4), no published theory supplies its two derived ingredients (Section 6.3), and the specification is therefore carried forward constrained, not realized. The second edition adds the derivation chain (Section 7), where the missing object is named in both halves; the final pass builds its matter half at the equation-of-motion level, closes its scalar lensing carrier constructively, and assembles the candidate action (Section 7.6) — a scaffold, gated, not validated.

2 The kernel and its empirical standing

2.1 The kernel and the status of its coefficient

The scale $c^2\sqrt{\Lambda}$ is forced by dimensional analysis once one decides the vacuum sets an acceleration; only the O(1) coefficient is a choice. The framework's 32π factors as $4\times8\pi$, where 8π is the GR coupling (the same 8π in $\rho_{\rm DE} = \Lambda c^2/8\pi G$) and the residual 4 is a motivated normalization posit, not a derived theorem. An adversarial audit of forcing routes (dimensional, holographic, CKN-EFT, GUT-ladder) returned four worked negatives: **no route forces the coefficient**, and the CKN channel in particular carries a free O(1) and makes no laboratory prediction (Blinov & Draper 2021, arXiv:2107.03530, find the generic g−2/Lamb-shift sensitivity ∼16 orders short of gravitationally motivated depletion). The honest summary, locked in the repository's convention file: the value is empirically excellent; the coefficient is a posit; the scaling $a_0 \propto \sqrt{\Lambda}$ is the physics.

2.2 Empirical standing at z = 0, and the convention-robustness rule

The kernel's working phenomenological layer is modified inertia from the de Sitter–Unruh temperature: a detector with proper acceleration a in de Sitter space sees $T_{\rm eff} = (\hbar/2\pi c k_B)\sqrt{a^2 + (cH)^2}$ (Deser & Levin 1997, arXiv:gr-qc/9706018), and the induced interpolation fits the SPARC radial-acceleration relation at **0.105 dex** scatter. Whether the framework's a_0 is "low" relative to SPARC turns out to be a fit-metric artifact, and the program treats this as a standing methodological rule: at fixed $\Upsilon = 0.70$ the RAR-preferred a_0 swings $\sim$40% with the fit metric (8.5×10^{-11} unweighted dex-scatter; 1.1×10^{-10} inverse-error-weighted; 1.3×10^{-10} linear). Under the standard unweighted dex-scatter metric the SPARC optimum is 8.48×10^{-11} and the framework value 9.36×10^{-11} is **within 0.3% of optimal scatter** — indistinguishable from rival normalizations within the M/L and metric systematics. An earlier in-program claim that the value was "$\sim$20% low" was exactly this artifact; it was caught and retracted, and every subsequent deficit claim in this paper has been checked against the same rule (alternative footing, alternative weighting, alternative Υ, alternative a_0(z) branch) before being reported.

The kernel's liability ledger at z = 0 rides along with every result below and is never dropped: galaxy clusters retain a $\sim$2× unseen-mass residual under any MOND-type law; the CMB third peak requires a covariant completion with a dark field; and the weak-lensing RAR early/late split (Section 5.2) is a standing exposure for any property-independent law, this one included.

2.3 The a_0(z) claim, stated with its conditionality string

The framework's distinctive cosmological claim is $a_0(\mathrm{z}) \propto \sqrt{\rho_{\rm DE}(z)}$. Three readings are distinguished and locked:

- **Constant** a_0 (the strict reading): the KMS/thermal argument strictly licenses only the asymptotic de Sitter fixed-point value. This is the fallback, and it is safe against all current data.

- **Declining** $a_0 \propto \sqrt{\rho_{\rm DE}(z)}$ (the framework's ansatz): with the DESI DR2 $w_0 w_a$ fit ($w_0 = -0.752$, $w_a = -0.86$) this is non-monotonic — a +6% bump at z ≈ 0.4, then 0.86× at z = 2 and 0.737× at z = 3 (the zero-lag value; with the response-gate selection the prediction is $\sim$0.65, and the full gate- and initial-condition-dependent band is [0.37, 1.0]). The extension is self-consistent only at z $\lesssim$ 1.3; at z = 3 it is a non-adiabatic extrapolation. A blind Unruh–DeWitt computation found that no standard state yields T $\propto \sqrt{\rho_{\rm DE}}$ in the matter-dominated era, so the declining branch is a *posited effective law whose standard-state derivation is computed-excluded*, to be decided empirically.

- **Rising** $a_0 \propto cH(z) \cdot E(z)$ (the rival, never the framework's law): excluded at $\Delta\chi^2 \approx 49$ on the clean kinematic data and CMB-excluded as modified inertia ($a_0(z_{\rm rec})$ would exceed the recombination bath acceleration by orders of magnitude, prescription-independently).

Two empirical updates are carried at full weight. First, the CMB does **not** discriminate declining from constant: with the bath (total real-space) acceleration as the μ-argument — the correct argument for a nonlinear, nonlocal law — the acoustic modes sit deep-Newtonian and both branches are CMB-safe; only rising dies there. Second, MUSE-DARK III (Ciocan, Bouché et al. 2026, arXiv:2604.22613) reports an apparent RAR-a_0 *rising* $\sim$2× by z ≈ 1. This is read as **contested and non-diagnostic**, for three verified reasons: ΛCDM-with-baryons simulations reproduce the same apparent rise (Magneticum: Mayer et al. 2023, arXiv:2206.04333); the same team's clean axis — the baryonic Tully–Fisher zero-point — is a null (±0.08 dex), internally contradicting a genuine 2× rise; and a rise of that magnitude is already excluded by Newtonian-regime high-z disks (Milgrom 2017, arXiv:1703.06110). The declining reading is therefore *weakened and contested, not falsified*; the only clean arbiter is the sign of the z ≈ 3 BTFR offset (Section 8).

3 The covariant-host trilemma: every published host fails a named gate

The decisive external gate for this section is the Cassini external-field-effect quadrupole. A MOND-type modification in the Galactic external field $g_{\rm ext} \approx 2.15\times10^{-10}$ m s^{-2} sources an anomalous quadrupole Q_2 in the outer solar system; the 2026 measurement is $Q_2 = (1.6 \pm 1.8)\times10^{-27}$ s^{-2} (Park, Hees, Famaey et al., arXiv:2602.17884), i.e. a 2σ acceptance window of $[-2.0, +5.2]\times10^{-27}$ s^{-2}. Desmond, Hees & Famaey (2024, MNRAS 530, 1781, arXiv:2401.04796) established the structural conflict: the SPARC RAR demands a gradual Newton–MOND transition where Cassini demands a sharp one, a 3–15σ class-level tension for AQUAL/QUMOND-type theories. The program independently re-derived and audited the quadrupole machinery (10-facet audit, including two owned corrections of earlier in-program misstatements) before applying it.

Row 1 — AeST (Skordis & Złośnik, PRL 127, 161302). CMB-safe, $c_{\rm GW} = c$, ghost-free — and Cassini-excluded robustly. The RAR-fitting quadrupole is $Q_2 \approx 3$–5×10^{-26} s^{-2}; at the framework's *own* lower a_0 the value is 3.2×10^{-26} — the footing suppression is real (×0.69) but acts on a ∼10× excess over the 2σ ceiling. The theory's one natural screening scale cannot help: the AeST mass term has $1/\mu \gtrsim 1$ Mpc (CMB/stability-pinned), so at 50 AU the screening term is ∼40 orders down and AeST reduces to pure QUMOND in the solar system. The surviving knob, β_0, is precisely the RAR-pinned transition sharpness, so the Desmond tension transplants intact.

Row 2 — single-scalar Galileon / k-mouflage hosts (Babichev, Deffayet & Esposito-Farèse 2011, arXiv:1106.2538, and the one-field DE+MOND construction). Cassini-safe by Vainshtein screening, but excluded structurally and data-independently: the MOND-required non-analytic kinetic function must straddle spacelike (in-galaxy) and timelike (cosmological) field gradients, so a **singular surface** on which the scalar ceases to propagate surrounds every galaxy (Bruneton & Esposito-Farèse, arXiv:gr-qc/0607055; restated arXiv:2503.11174). The side exits are closed: the construction's curvature-coupled L_4/L_5 screen is the GW170817-killed class; the flat cubic screen is ISW-excluded at 7.8σ; and the DESI-favored Galileon is a different operator (an earlier in-program conflation, withdrawn). Building "AeST plus a Galileon screen" reinstates the Row-1 Cassini bill.

Row 3 — the Deffayet–Woodard nonlocal-metric class (Deffayet, Esposito-Farèse & Woodard 2011, arXiv:1106.4984; Deffayet & Woodard, JCAP 04 (2026) 081, arXiv:2512.10513). This row is the paper's principal new covariant-sector computation, and its history is reported honestly: an earlier in-program statement that the perimeter was closed was **amended** when this class was found alive with its decisive gates uncomputed — a real retreat, recorded at full weight — and the gate was then computed the same day. The result re-closes the perimeter:

- *The static limit was derived* (two independent routes agree): the model's static weak-field equations are exactly Bekenstein–Milgrom AQUAL in the total potential, with

$\mu(g) = 1 - e^{-y}(1 - y/2)$, $y = 2g/3a_0$, and $\Phi = -\Psi$ enforced (correct lensing). The DEW papers themselves name the external-field question "a matter for future study"; to our knowledge this is its first computation. One structural feature matters: their published $f(Z)$ makes μ overshoot unity ($\mu_{\rm max} \approx 1.025$ at g = $4.5a_0$) — an anti-MOND lobe that mixes the sign of the EFE source.

- *The verdict:* with the repository-verified Desmond formalism plus a direct nonlinear AQUAL solver (validated against the published function-space anchors and stable to 0.3% under grid refinement), the published 2026 function gives $Q_2 = \mathbf{+2.80}{\times}10^{-26}\ \mathrm{s}^{-2}$ **(14.6σ) at the model's own $a_0 = \mathbf{1.2}{\times}10^{-10}$**, and **$+\mathbf{1.74}{\times}10^{-26}$ (8.8σ) at the framework footing**; across the pre-registered $g_{\rm ext}$ robustness range it never drops below 3.2σ. The model's *other* published function (the 2011 "example 2") is also excluded, at 4.1σ with the opposite sign.

- *The rescue scan (the both-ways obligation):* an extended attempt was made to make the class pass, scanning a function family that preserves the pinned deep-MOND coefficients. Passing members exist — but every one passes only through an accidental sign-cancellation of the anti-MOND lobe, is **$g_{\rm ext}$-fragile** (pass bands $\sim \pm 0.15{\times}10^{-10}$, narrower than the $g_{\rm ext}$ uncertainty, between opposite-signed failures), pays an **RAR penalty** of 2.4–3.4× the achievable binned-χ^2 floor even at its own best-fit Υ, and is **theory-unselected** (nothing in the construction picks the cancellation point). The strongest single rescue cell found anywhere in the scan is reported at full weight — and it fails at both ends of the $g_{\rm ext}$ range, pays the family's RAR penalty in absolute terms, and requires the framework's footing rather than the model's own. It is a tuning sliver, not a viable published host.

Row 4 — modified inertia (Milgrom 1994, arXiv:astro-ph/9303012; 1999, arXiv:astro-ph/9805346). The framework's own conceptual basis, and the only class that evades the Cassini quadrupole *by class* (the constraint bounds modified gravity; a trajectory-dependent law sources no static Q_2 analogue, and the repository's own F4 Saturn computation confirms the evasion quantitatively). Its gate is different in kind: **no covariant completion exists.** Milgrom's theorem shows a local Galilei-invariant modified-inertia action with both limits is impossible, while explicitly licensing nonlocal worldline functionals; "preliminary suggestions in the non-relativistic regime" remains the literature's full inventory.

The honest perimeter statement. Every published covariant host now either fails a computed data gate (Rows 1–3) or does not exist as a covariant theory (Row 4). The Row-3 closure is a Desmond-type function-space incompatibility at the 3–15σ class level, not a theorem; its residue is the named tuning sliver, stated above rather than hidden. The perimeter points every arrow at one target: a covariant (necessarily trajectory-nonlocal) completion of modified inertia — and Sections 4–6 are the program's confrontation of that target with exact computation and data.

4 The modified-inertia program: selection, kills, and exact no-go results

This section reports the worldline-level campaign in the order it was run. Its net result is framework-unfavorable on mechanism and solar-system embedding, framework-favorable on one structural opening, and it ends in a specification, not a construction. Two normalizations are carried throughout, per the program's convention rule: the framework's s $= a_0 = 9.36{\times}10^{-11}$ and the hostile bath-natural s $= cH_\Lambda = 5.42{\times}10^{-10}\ \mathrm{m\,s^{-2}}$.

4.1 F1 = Milgrom (1999) is excluded by planetary ephemerides by ×54,000

The natural bath-inertia ansatz — inertia proportional to the Deser–Levin response *excess* over the de Sitter floor, $\mu(x) = [\sqrt{x^2+1} - 1]/x$, $x = a/cH$ — is algebraically identical (verified by substitution) to the published Milgrom 1999 proposal (arXiv:astro-ph/9805346). Its deep-MOND limit emerges for free, with coefficient $a_0 = 2cH$ (11.6× = 2Z above the SPARC-fitting value — the bath's own coefficient is on record and does not fit galaxies either). Its high-acceleration tail is fatal: $\mu \approx 1 - cH/a$ implies a *constant* anomalous acceleration of magnitude cH $\approx$ 5.4–6.6$\times 10^{-10}$ m s^{-2} on every solar-system body. Against the Cassini-era Saturn bound (anomalous radial acceleration $< 10^{-14}$ m s^{-2}; Folkner, via arXiv:1001.3686 §VI) this is an excess of **$\times$54,000 (ρ_{DE} footing) to $\times$65,000 (ρ_{total} footing)** — the kill is footing-robust, and it survives even the soft planet-wide INPOP bound (Fienga et al., arXiv:0906.3962) by $\times$2.5. The same arithmetic kills the McCulloch "quantized inertia" proposal (arXiv:astro-ph/0612599), whose kernel is the same difference family ($\times$54,600–68,000, both footings, machine-checked). Historical fairness is recorded: in 1999 a $\sim$cH constant anomaly was arguably supported by the then-unexplained Pioneer anomaly; the planetary bounds that kill it came later. To our knowledge the ephemeris exclusion had not previously been applied to this specific proposal.

The viability line this kill draws is sharp and useful: the ephemeris bound falls cleanly *between* the linear tail (dead $\times$54,000) and a quadratic tail $\mu - 1 = O((cH/a)^2)$, which yields $\delta a(\text{Saturn}) = 2.3 \times 10^{-15}$ m s^{-2} and is safe with a $\times$4 margin. Any viable kernel must close at least one order faster than the natural one.

4.2 The see-saw no-go, and the selection of F4

Within the bath-response family (the only live mechanism class), a scan over couplings (sympy- and numerically verified) found a **mutual-exclusion no-go in the difference family**: a spectral gap large enough to exponentially suppress the solar-system tail ($\varepsilon \gtrsim 13.5$) destroys the deep-MOND limit, and restoring the deep limit re-fattens the tail worse than the gapless case. The see-saw has no crossing point. Exactly one coupling inside the family escapes: differentiating instead of subtracting — inertia as the *susceptibility* of the detector's thermal state to acceleration, $m_{\mathrm{eff}} \propto dT_{\mathrm{eff}}/da$ — which yields exactly the standard MOND interpolation function $\mu(x) = x/\sqrt{1+x^2}$ with no free parameter, a quadratic (Saturn-safe, $\times$4) tail, and the deep limit intact. This coupling, labeled **F4**, was *selected* by the pre-registered spec sheet; it was not assumed, and it was never claimed as derived. Its implied coefficient is $a_0 = cH_\Lambda$ — still off by exactly Z, consistent with the banked verdict that Z is data-selected. The susceptibility form does not appear in Milgrom 1999, and a deep literature search (Section 4.5, agent N4) confirmed it is unpublished — recorded as a priority absence, with the graveyard prior stated alongside.

F4's five passes, each pre-registered: (i) the deep-MOND limit, analytic; (ii) Saturn, $\times$4 margin where the published rival fails $\times$54,000; (iii) the SPARC RAR shape test on 175 galaxies — best unweighted scatter 0.1984 dex versus the McGaugh function's 0.1950 at the framework a_0 (+0.0034, under the +0.01 survival line) with physical Υ_{disk}; a hostile Υ-convention check then showed the ranking flips at the MOND-default $\Upsilon = 0.5$, so the convention-robust statement is that **SPARC cannot presently rank the two shapes** — F4 is competitive, not victorious; (iv) wide-binary/EFE consistency (Section 5.1); and (v) a predictive inversion of the Gaia DR4 fork: F4's sharp knee yields a $\sim$+4% flat deep-bin velocity-boost plateau versus +13–16% for soft MOND shapes, so a clean DR4 null at $\sim$3% sensitivity kills soft shapes while F4 survives, and a +10–15% detection kills F4 (both kill conditions pre-registered before DR4).

4.3 The kills

4.3.1 Eccentric orbits (Door IVa). The non-circular modified-inertia correction was derived analytically two ways and integrator-verified: per-orbit perihelion drift $\Delta\varpi = -\pi(4 + e^2)\sqrt{1-e^2}\cdot s^2 a^4/2(GM)^2$, retrograde and $\propto a^4$, so Saturn binds rather than Mercury. F4 passes — but only at 2σ under the hostile normalization (Saturn -0.307 mas/cy versus INPOP15a-C2's 0.05 ± 0.20), with a $\times$25 spread across published ephemeris bounds, and the result is *prescription-laden*: 100% of the signal vanishes under orbit-averaging. The same analysis opened a sharper channel: the Sun's own modified reflex.

4.3.2 The solar reflex kills the instantaneous worldline reading at both normalizations (Door IVb). The Sun's barycentric acceleration (Jupiter-driven, $|a_\odot| \approx 2.1\times 10^{-7}$ m s^{-2})

is quasi-steady, so its anomalous F4 response does *not* orbit-average away. A full nonlinear Levenberg–Marquardt ephemeris-fit emulation (six bodies, 35-year arc, partials re-integrated per iteration; the agent caught and superseded its own linearized version as non-verdict-grade) gives: hostile normalization — post-fit Mars residual 377 m against 1.5 m ranging accuracy ($\times$251); framework normalization — 12.7 m ($\times$8.5, Mars-carried). The unabsorbable carrier is the Earth–Jupiter synodic (1.09 yr) response to the Sun's anomalous reflex, anti-correlated with r_J^2; fits that absorb it into GM_J require $|\delta \ln GM_J|$ 20–1000$\times$ the Juno determination's uncertainty (Durante et al. 2020, GRL 47), which independently refutes the absorption. The survival line is $s < (0.34\text{–}0.40) \cdot a_0$: **both candidate normalizations sit above it. Instantaneous, center-of-mass-worldline F4 is dead in the solar system.** The constituent-acceleration escape (evaluating μ on stellar-interior accelerations) would switch off galactic MOND for stars and is incoherent with the phenomenology.

4.3.3 The mechanism is refuted at all orders (Doors I and I-b). At order λ^2, machine-verified exact identities force every Unruh–DeWitt worldline response coefficient into the form $A(\kappa) + a^2 B(\kappa)$ with $\kappa = \sqrt{a^2 + H^2}$-only coefficients; matching the F4 susceptibility requires a kernel with a pole exactly at the deep-MOND point, which is impossible (sympy-exact ODE no-go). What the conformal bath actually delivers is thermal-mass *dressing* — the anti-MOND direction (inertia added at low a, where MOND needs a deficit). The named non-perturbative loophole was then closed by the exactly solvable Gaussian (harmonic UDW) detector: the field commutator pulled back on *any* stationary Deser–Levin worldline is the universal contact distribution $(i/2\pi)\,\delta'(s)$ — the dissipation kernel is local and trajectory-blind at every coupling strength — so the exact resummed response keeps the $A(\kappa) + a^2 B(\kappa)$ census form at all orders. Numerically, the exact dressing G is positive (inertia added) everywhere probed across five decades of coupling, including the gapless ultra-strong corner that a pre-run guess had hoped could go negative (recorded); the best achievable approach to the F4 shape, even with free affine parameters, misses by 25% of range. The "perturbation theory must fail at vanishing inertia" intuition was tested and is false: the exact theory is smooth at a $\to 0$ and the inertia simply does not vanish. The raw coefficients produced ($\lambda^2/8\pi$; $\kappa^2/48\pi^2\Omega^2$; $1/4\pi^2$; $1/6$; $1/2\pi^2$) are reported in isolation per the program's discipline: none is Z-related, and the 1/6-versus-1/Z near-miss (3.5%) is flagged as structurally meaningless. **F4's worldline-bath mechanism candidacy — perturbative and non-perturbative — is closed.** The lemma names its own boundary: the closure rests on the Huygens property of the conformal field; the unique field-side escape is fields whose Green functions have tails.

4.3.4 The laboratory walls. A confrontation with published laboratory data (all four prescription readings $\times$ both normalizations) found: the banked background-acceleration readings are immune by 10^{12}–10^{19} in every published experiment (every laboratory background trajectory has x $\gg$ 1); the amplitude/per-axis prescription is **laboratory-dead** — Gundlach et al. (PRL 98, 150801 (2007)) exclude it by $\times$148–1116 (framework) to $\times$855–6464 (hostile) in the kernel scale, with the predicted oscillator-period collapse integrator-verified against the paper's own simulation; MICROSCOPE's $\eta = [-1.5 \pm 2.3 \pm 1.5] \times 10^{-15}$ (arXiv:2209.15487) is structurally blind to universal modified inertia ($\eta_{\mathrm{F4}} = 0$ exactly); and one genuine door is open with existing instruments (Section 8, the atom-interferometry protocol). The Gundlach wall is added to the spec sheet: any future mechanism must deliver background-anchored, not amplitude-keyed, linear laboratory response.

4.4 The non-Huygens door: exact results

After 4.3.3, the unique bath-side escape was fields with retarded tails (massive scalars; the minimally coupled massless scalar in dS). A five-agent campaign mapped it exactly. The results below are, to our knowledge, new.

N1 — the closed-form commutator: trajectory-blindness genuinely breaks. For the massive scalar in dS_4 (Bunch–Davies, both principal and complementary series; effective mass $M^2 = m^2 + 12\xi H^2$, $\nu = \sqrt{9/4 - M^2/H^2}$), the field commutator pulled back on the stationary Deser–Levin family (proper acceleration a, $\kappa = \sqrt{a^2 + H^2}$, $\beta = H^2/\kappa^2$) is, exactly:

$$C(s) = (i/2\pi)\,\delta'(s) + (i/8\pi)(M^2 - 2H^2)\cdot\mathrm{sgn}(s)\cdot{}_2F_1(3/2+\nu,\ 3/2-\nu;\ 2;\ -(H^2/\kappa^2)\sinh^2(\kappa|s|/2)).$$

The contact term is universal (the local part of dissipation stays trajectory-blind); the tail is an irreducibly **two-variable** function of $(\kappa, \beta) \longleftrightarrow (a, H)$ for every $M^2 \notin \{0, 2H^2\}$. The conformal point $M^2 = 2H^2$ is precisely where the tail coefficient vanishes — the 4.3.3 lemma is the measure-zero special case — and the minimally coupled massless limit is the clean boundary case C $= (i/2\pi)\delta'(s) - (iH^2/4\pi)\,\mathrm{sgn}(s)$ (H-aware, a-blind; the Allen IR pathology is quarantined in the noise sector). KMS thermality at $T = \kappa/2\pi$ survives for every mass (verified to 10^{-25}): the *temperature* stays κ-only; it is the spectral content that breaks. At fixed κ the tail's first coefficient runs with a^2 at the exact mass-independent rate $1/4\pi$ and flips sign across the family. Consequences, both ways: the all-orders census of 4.3.3 does **not** extend to non-Huygens fields, and the kinematic exclusion of the F4 shape dissolves ($\mu_{\mathrm{F4}} = \sqrt{1-\beta}$ is smooth precisely where the κ-only obstruction had a pole); but nothing here shows any mass actually *produces* the MOND response — the impossibility proof is removed, not the burden of construction.

N2 — the memory Langevin equation, the universal gate, and the knee window. Integrating out a tail field exactly gives a quantum Langevin equation whose dissipation kernel is a functional of the trajectory segment. Two hard structural results follow. (i) **The adiabatic limit localizes**: below the kernel's knee frequency the memory force renormalizes into ordinary local (a, H, m)-dependent inertia — $m_R(\Omega) - m_R(0) \propto \Omega^2$ with no $|\Omega|$ term — so memory cannot generate the MOND shape in-band; the shape and sign must come from the adiabatic tail amplitude (N3's lane). (ii) For rotating-acceleration (helical) worldlines the reactive response is suppressed above the knee as $m_R(\Omega) \to M_0/\Omega^2$ — **p = 2, universally** (sharp or smooth kernel), with the quadrature channel at p = 3. The Sun's reflex runs at $\Omega \sim 10^{-8}$ s^{-1} and galactic orbits at $\sim 10^{-16}$–10^{-15} s^{-1} — an eight-decade lever — so the solar suppression that Door IVb demands (×8.5 framework, ×285 hostile) is available, and the knee window is **non-empty by 4.3–5.1 decades:** $\varkappa \in$ **[2.0**$\times10^{-14}$**, 2.5**$\times10^{-9}$**]** s^{-1}, robust to footing and normalization. As a field mass this is $mc^2 \in$ **[1.3**$\times10^{-29}$**, 1.6**$\times10^{-24}$**] eV** — the fuzzy-dark-matter band. The framework-unfavorable half is decisive for the program's self-understanding: **the no-new-scale pure-dS knee ($\varkappa \sim$ H) is excluded three independent ways** — it would tilt the effective a_0 by ∼4.9 dex across the RAR band (observed total scatter 0.057–0.11 dex), suppress the band absolutely by $\times 3\times10^5$, and make clusters respond $\sim 10^3\times$ more strongly than galaxies. The memory scale the dS bath supplies for free does not do the job; the knee must be a *new* scale, 4–9 decades above H.

N3 — the MOND-signed channel, and the four walls that close the worldline. The massive-scalar tail dressing of a worldline-coupled charge was computed exactly (calibrated against Burko–Harte–Poisson, arXiv:gr-qc/0201020, and Haas–Poisson, arXiv:gr-qc/0411108). The framework-favorable finding, stated at full weight: for all $m^2 < 2H^2$ the induced inertia change is a **deficit, larger at low acceleration — the MOND sign — the first and only right-signed bath channel the program has found** (every Huygens-sector channel was anti-MOND). The framework-unfavorable finding, equally weighted: the window is **empty**, four independent ways, with margins immune to every convention axis. (1) The coupling wall: the bath carries energy scale H per coupling while the inertia to modify is nucleon-scale — the hierarchy $(m_p/M_{\mathrm{Pl}})(H/M_{\mathrm{Pl}}) \approx 1.9\times10^{-79}$ means the galactic job needs a per-nucleon scalar charge $\sim 10^{42}$ times the Cassini-allowed universal coupling (a fifth force $\sim 10^{81}\times$ gravity); at allowed couplings the effect is 10^{-86} of the job. (2) The shape wall: the dressing's high-a tail is $(\ln x)/x$ at every mass — the dead F1 ephemeris class, softened only by the log; the pre-registered hope of a mass-tunable falloff exponent was refuted by the computation itself (recorded as a self-caught correction). (3) The frequency-bracketed variant is solar-safe but source-impossible by ≥87 orders. (4) The equilibration cap: the light-field IR enhancement $\langle\varphi^2\rangle = 3H^4/8\pi^2m^2$ (Starobinsky–Yokoyama, arXiv:astro-ph/9407016) needs $1.5(H/m)^2$ Hubble times to build; our Λ-phase is 0.2–0.35 H^{-1} old, so it never materializes, and the $m \to 0$ limit is the published mass-loss runaway, not MOND. One line: **the dS bath has the right length scale and the wrong energy scale** — $\kappa = \sqrt{a^2 + (cH)^2}$ bends at cH, so any dS-bath object knows *where* a_0 is, but $a_0 \;\propto\; \sqrt{\rho_{\rm DE}}$ can be a kernel of the law, not the strength of a worldline force.

N4 — the published map. The door's three ingredients all exist in print and have never been combined: Takagi's massive-field worldline kernels carry (a, H) beyond κ; Haas–Poisson prove dS tails do not renormalize away (the inertial mass of a charge in FLRW is an exact, non-localizing history integral); and Milgrom's time-nonlocal modified-inertia models (arXiv:2208.07073) supply the explicit WEP-safe target functional. No published work connects tails to MOND (priority opportunity and graveyard prior both recorded). The published magnitude record is uniformly hostile: every computed tail force scales as q^2/m (25–80 orders short, with a 57-order composition spread that maximally violates universality), so the mechanism — if any — must make inertia multiplicative and clock-keyed, for which no computed exemplar exists in either direction.

N5 — the data discriminate keying. Pure frequency-keyed inertia, the naive evasion of the solar-reflex kill, is **SPARC-dead**: best variant +0.0226 dex past the pre-registered death line, failing with the predicted $a = \Omega \cdot V$ structure (the per-galaxy residual trend against $V_{\rm flat}$ flips sign at 4.4–5.2σ; one fitted Ω_0 implies a ×15 spread in the effective acceleration scale where the data demand one). But the kill does not extend to frequency-*dressed* acceleration-keying: a single dressing $a_{0,\rm eff} \propto [(1 + \Omega/H_0)]^{-p}$ with **p ∈ [0.069, ≥1] keeps SPARC within +0.010 dex (mild p ≈ 0.2–0.5 costs nothing) while suppressing the solar reflex by ≥3×10⁷** — the first surviving evasion of Door IVb inside the modified-inertia class. The corridor's price is itself falsifiable: corridor members suppress the wide-binary boost (fully off for p ≥ 0.5) and kill all laboratory signatures. A context row reframes Door IVb: the exponentially-saturating McGaugh RAR shape passes the solar budget *undressed* — what died at IVb was F4's power-law tail, not acceleration-keying itself.

Closure of the worldline class. N3's magnitude walls plus N2's no-new-scale exclusion close every bath-coupled light-scalar worldline realization; the Huygens kill stands beneath them at all orders. **No bath-coupled point worldline carries MOND.** What the door yields is not a mechanism but the sharpest specification the missing object has ever had — Section 6.

5 The data program

5.1 Wide binaries: the observable is degeneracy-limited

On the same Gaia catalog, Chae (2023, ApJ 952, 128, arXiv:2305.04613) reports a $\sim 5\sigma$ low-acceleration velocity boost and Banik et al. (2024, MNRAS 527, 4573, arXiv:2311.03436) a $\sim 16\sigma$ Newtonian null. The program replicated both teams' selections on the El-Badry, Rix & Heintz (2021, MNRAS 506, 2269) eDR3 catalog, with pre-registration before every run, and produced a methodology result addressed to both sides:

- **Model-free diagnostics.** The scale-free core of the observable ($\tilde{v} = v_{\rm sky}/\sqrt{GM/r_{\rm sky}}$ depends only on eccentricity, phase, and viewing geometry) implies a Newtonian population predicts a *flat* median across acceleration bins; any deep-bin rise must be noise, contamination, eccentricity structure, or new physics. The noise-folding diagnostic D1 (the 2D vector fold inflates the deep-bin median preferentially; scalar noise models miss this) is ≈ 0.12–0.15 on the faithful samples — the same order as the +2–11% boost being sought. The escape-bound diagnostic D2 is theory-laden: pairs with $\sqrt{2} < \tilde{v} < 1.65$ are contamination under Newton but a boosted bound population under EFE-suppressed

MOND — the same pairs, classified oppositely by hypothesis. The mass-sensitivity diagnostic D4 is clean (1.3–1.4×, below alarm). One real bug (a mass-cubic inversion outside its valid window, which had produced a spurious ∼0.48 super-escape) was caught, owned, and retracted mid-program; the faithful super-escape is ≈ 0.10 for both teams' cuts.

- **The verdict.** Calibrated on the high-acceleration anchor, the deep bins sit $\sim 3\sigma$ above the flat-contamination Newtonian baseline, and that excess is *fully absorbed* by a separation-dependent triple fraction (∼0.16) that independently matches the measured super-escape. The deepest-bin residual is 1.7–2.4σ standalone, not significant. A real modified-gravity orbit integration further showed the predicted deep-bin statistic is sensitive to the unconstrained 3D orbital (a, e) prior — a finding that **weakens everyone's significances**, the published $\sim 16\sigma$ null and $\sim 5\sigma$ boost alike. The diagnostics are insensitive to the choice of a_0 (binning-scale check: median $\tilde{v}$ unchanged at 0.661). Bottom line, carried into the falsification matrix: **wide binaries are currently null-informative / degeneracy-limited at the framework's a_0 — neither strike nor support — awaiting Gaia DR4** (line-of-sight velocities break the boost↔contamination degeneracy and pin the orbital prior).

- **The DR4 fork, sharpened by F4.** Under the vector modified-inertia prescription at the framework footing, the deep-bin velocity boost is ∼+4% (flat plateau) for the F4 shape versus +13–16% (simple) and +11–14% (McGaugh RAR) for soft shapes. The fork is pre-registered: a clean DR4 null at ∼3% sensitivity kills soft MOND shapes while the F4 shape survives at/below sensitivity; a +10–15% detection kills the F4 shape while confirming soft shapes; only ∼2%-level precision tests the F4 shape itself.

5.2 The lensing RAR early/late split: the strongest standing exposure, and it hardens

Brouwer et al. (2021, A&A, arXiv:2106.11677) measured the weak-lensing RAR from KiDS-1000, finding early-type galaxies offset above late types at fixed $g_{\rm bar}$ — a direct challenge to any property-independent law, this framework's included, since a universal force or clock law is type-blind. The program audited this exposure with inverted hostility (the pre-registration treats framework-favorable outcomes as the ones owing hostile review):

- **Replication.** On Brouwer's released ESD profiles with the full 30×30 covariance: χ^2 = 119.9/15 dof → 8.8σ, early above late in 15/15 bins, mean offset +0.261 dex in $g_{\rm obs}$ — validating and slightly sharpening the published "$\geq 6\sigma$". The split is a_0-independent (type-dependence at fixed $g_{\rm bar}$ involves no acceleration-scale freedom).

- **The conversion axis, measured rather than bracketed.** The released conversion $g_{\rm obs} = 4G\Delta\Sigma$ assumes an SIS profile; the true factor C is profile-dependent, and a generic concentration bracket initially suggested the split could erode to 5.0σ (u−r) / 2.1σ (Sérsic). That bracket was then **superseded by measurement**: building per-class, per-bin conversion factors from the actual lens populations (9 SHMR × c(M,z) configurations; Moster+13 arXiv:1205.5807 / Behroozi+13 arXiv:1207.6105; Duffy+08 arXiv:0804.2486 / Dutton & Macciò 14 arXiv:1402.7073; the Mandelbaum+16 arXiv:1509.06762 type-dependent halo-mass offset), the type-differential is $\Delta \log C \approx +0.001$ dex — two orders of magnitude smaller than the bracket and *opposite-signed* (split-strengthening in 14/15 bins), because at fixed $g_{\rm bar}$ early types are probed at larger radii (×1.52) but live in halos with proportionally larger scale radii (×1.62). **Refined survival:** 8.6–9.2σ **(u−r) and** 5.6–6.3σ **(Sérsic)**; a hostile probe shows even an indefensible ×6 early-only concentration boost leaves 7.6σ. The two morphology axes now corroborate rather than moderate each other.

- **The gas escape is disfavoured.** Closing the split by adding circumgalactic baryons to early types requires a differential $M_{\text{gas}}/M_*\ \approx 1.1$ — and eROSITA stacking (Zhang et al. 2025, arXiv:2411.19945) finds no hot-CGM differential between quiescent and star-forming galaxies below $\log M_* = 11$, exactly the relevant regime. The hedge that survives (a cool/warm-phase differential) is named but not independently motivated.

- **Independent re-measurement.** A from-scratch KiDS-1000 $\times$ GAMA re-analysis (181,477 isolated lenses, 21.3M sources; two real pipeline bugs — a units chain and a rotation-convention sign — were caught by pre-registered validation gates before any significance was computed) reproduces the released early-type ESD profile at the **$\sim$5–10% level** (median ratio 1.04, 13/15 bins within $\pm$30%, most within $\pm$10%); the late-type (weak-signal) class tracks the released profile (median 1.06) with larger per-bin scatter, and no independent σ is quoted until its error bars pass the gate. **Completed (agentK, 50-patch spatial jackknife, Hartlap-corrected):** the validation gate passes BOTH classes (early $\chi^2_{\text{fullcov}} = 11.9/15 \rightarrow\ 0.4\sigma$; late $19.6/15 \rightarrow\ 1.3\sigma$ — the late-class scatter was shape noise), and the re-measurement yields an **own-data early/late split of** 6.8σ (Hartlap-corrected headline; 9.1σ raw; +0.209 dex, early above late in 14/15 bins), independently confirming the released-profile 8.8σ with all four expected dilutions (lens count $\times$0.70, jackknife-vs-analytic covariance, the colour-split convention, the count-calibrated isolation window) named and quantified (`reviews/lensing_rar/agentK_remeasure_errors.md`). The released profiles on which the headline numbers rest are therefore independently validated in the strong-signal class.

- **The metric-passive wall (consequential for Section 4).** The same released data exclude *metric-passive* modified inertia — any theory in which lensing responds to baryons alone — at 40.5σ: the deep-bin lensing amplitude exceeds the baryon-only prediction by $\sim$230$\times$, unbridgeable by the conversion-factor bracket (0.79–1.19$\times$). Any surviving matter-sector law must be accompanied by a lensing-carrying partner.

- **Locked wording.** This is an exposure to the framework and to every property-independent modified-gravity/inertia law; it is *not* “ΛCDM confirmed” — Brouwer’s own comparison simulations (MICE vs BAHAMAS) disagree with each other about the split, so no specific dark-matter model is validated by it. The split is real at 8.6–9.2σ, survives its most plausible systematic at measured amplitude, and its named escape is disfavoured by post-2021 X-ray data. It is the program’s most significant framework-unfavorable result and is recorded as such.

6 The specification of the missing object

Every front above converges on one residue. The covariant perimeter (Section 3) leaves only a trajectory-nonlocal matter sector; the solar-reflex kill (4.3.2) requires the nonlocality (Milgrom’s theorem always required it; the instantaneous reading is now also data-dead); the lensing wall (5.2) requires a metric partner; and the non-Huygens campaign (4.4) fixes the sign, the keying, and the scale. The missing object is a **field-level hybrid** whose matter sector must reproduce, from the program’s own numbers:

1. **Sign:** an inertia *deficit* growing toward low acceleration — the $m^2 < 2H^2$ tail channel’s structure (N3), the program’s only right-signed channel;

2. **Keying:** acceleration-keyed in the galactic regime (pure frequency-keying is SPARC-dead, N5), with frequency dressing confined to the corridor $\mathrm{p} \in [0.07,\ 1]$ as a_0-dressing; the natural reactive $\mathrm{p} = 2$ of memory kernels sits solar-safe;

3. **A knee scale:** an ultralight field with $mc^2 \in$ [**1.3**$\times10^{-29}$**, 1.6**$\times10^{-24}$] **eV** (the fuzzy-DM band) — the one new scale the dS kernel cannot supply itself (N2's no-new-scale exclusion); the Gaia DR4 wide-binary amplitude is the knee-position discriminator (split at $\sim5\times10^{-28}$ eV);

4. **The length scale for free:** $\kappa = \sqrt{a^2 + (cH)^2}$ bends at cH — the bath kernel knows *where* a_0 is; what it lacks is the energy scale (N3's verdict line);

5. **The lensing partner:** required independently by the 40.5σ metric-passive wall; and the type-blindness liability (the $\sim 9\sigma$ split, Section 5.2) rides along unchanged as the hardest open phenomenological question for *any* universal law.

Sections 6.1–6.5 then confront this specification in order with the structure-formation ledger, the published field-level literature, the program's own first construction attempt, and the banked kill-test gauntlet run on the nearest published cousin. The confrontation ends in a pre-assembly kill (Section 6.4): the specification is carried forward constrained, not realized. Section 7 (second edition plus final pass; authoritative where the two overlap) carries the specification to its current state: the matter sector named and built at the EOM level, the partner narrowed to a unique class by theorem with its scalar sector closed constructively.

6.1 The knee band against published structure-formation walls

A hostile wall-map (every claim arXiv-pinned) of the knee band returns a clean two-regime verdict. **As dominant dark matter the entire band is dead several times over**: Lyman-α (m $> 2\times10^{-20}$ eV; Rogers & Peiris, arXiv:2007.12705), ultra-faint dwarf heating (m $> 3\times10^{-19}$ eV; Dalal & Kravtsov, arXiv:2203.05750), Milky Way satellites (m $> 2.9\times10^{-21}$ eV; Nadler et al., arXiv:2008.00022), CMB+LSS (dominant ULDM excluded outright for m $\leq 10^{-25.5}$ eV; Hložek et al., arXiv:1410.2896, 1708.05681), and PTA (EPTA DR2, arXiv:2306.16228) — so the hybrid *must* keep the density/lensing budget on a separate component; this is forced, not optional. **As a subdominant memory-carrier the band is open**, with a fraction-versus-mass ledger: f $\lesssim$ 1–3% is open across the *entire* band (the hardest pinch, f $\lesssim$ 1.3%, sits near 10^{-27} eV — eROSITA clusters, arXiv:2502.03353, with BOSS, arXiv:2104.07802); f $\sim$ 0.1 is open only above $\sim 10^{-26}$ eV (contested at 10^{-28}–10^{-25} eV, where a claimed S8-improving window, arXiv:2301.08361, is squeezed by the 2025 eROSITA result); f $\sim$ 0.2–0.3 only in the top two decades (Lyman-α fractional ceiling, arXiv:1708.00015). Black-hole superradiance — the only fraction-independent probe — has **no exclusion island anywhere in the band** (lowest scalar island floor 2.5×10^{-21} eV, 3.2 decades above the band top; Stott, arXiv:2009.07206). The Gaia DR4 knee-discriminator mass $\sim5\times10^{-28}$ eV sits at the hardest-pinched part of the ledger, where the field's de Broglie wavelength ($\sim$19 Mpc) makes it a quasi-homogeneous background; **the decisive quantitative question — pre-registered when this wall-map was banked — was whether the mechanism works at f $\lesssim$ 3%** (e.g. because the deficit amplitude is keyed by the bath kernel while only the knee position depends on the carrier density). The registered fork read: if it requires f $>$ 3% at that mass, the band's lower 3.5 decades close and the knee is forced into [$10^{-25.5}$, 1.6×10^{-24}] eV. **Update at integration: that calculation has now been run (Section 6.4) and decides negative on both branches of the fork** — no decade of the band passes the amplitude requirement, so the band does not narrow toward the top decades; for any build in which the carrier itself pays the MOND amplitude, it closes.

6.2 The superfluid-class convergence

The nearest published cousin of the required architecture is Berezhiani–Khoury superfluid dark matter (arXiv:1507.01019). Its hostile literature map returns **WOUNDED, not dead**: the one-field realization is squeezed by three independent published walls — the weak-lensing RAR staying MOND-flat to $\sim$1 Mpc (Mistele et al., arXiv:2303.08560, hardened by arXiv:2406.09685), Milky Way vertical kinematics (arXiv:1812.08169, 1911.12365), and the phonon's internally inconsistent double role plus stellar Cherenkov losses (arXiv:2009.03003, 2103.16954) — while the class remains alive as a program (Physics Reports review, arXiv:2505.23900, Feb 2026) with genuine points in its favor (strong-lensing fits, arXiv:1809.00840; fast-bar friction). The instructive fact is that the class's own critics drive it toward exactly the spec's architecture: the published fixes are a **two-field split** — a separate memory/force carrier and a separate density/lensing carrier (arXiv:2208.14308, 2009.03003) — which is the Section-6 configuration arrived at independently from the program's walls. A priority search confirms that the specific combination the spec requires — a dS/Unruh bath-keyed κ-kernel carried by a superfluid/ultralight sector — **does not exist in the published literature** (the nearest items: Berezhiani–Khoury's own $a_0 \sim H_0$ coincidence remark, explicitly not a bath derivation; Volovik's dS thermal bath without MOND, arXiv:2410.04392). The territory is unclaimed in the published literature in both directions: nobody has built it, and nobody outside this program has killed it. (That sentence was banked before the construction attempt; Section 6.4 has since supplied the program's own kill of the fraction-limited version of exactly this combination, so the unclaimed-territory statement now applies only to variants whose amplitude is paid outside the carrier's energy budget.)

6.3 Candidate-matrix verdict: the field is empty

The eight-point spec was operationalized as a *gated* scoring matrix (items 4–7 are pass/fail data gates; items 1–3 are the structural ingredients derived above; item 8 is the type-split differentiator) and run against every published field-level candidate that could host the missing object: the superfluid family (vanilla Berezhiani–Khoury; Khoury's "Another path," arXiv:1602.05961; the two-field split), baryon-interacting DM (arXiv:1712.01316), pure fuzzy/ULDM, the 2026 baryon-correlated DM EFT (arXiv:2605.20217), dipolar DM, and emergent gravity, with AeST and DEW as calibration rows. The verdict: **the field is empty at the spec's standard.** The best live score is 3/8 (BIDM — whose firm YESes are the cells that pass because nothing new happens in the solar system, and whose identity is hostile to the kernel: a_0 emerges there from interaction microphysics, not cosmology). Vanilla Berezhiani–Khoury scores 2/8, with lensing a published NO *on the program's own data axis* (the SfDM-vs-Brouwer confrontation, arXiv:2303.08560: closest-match $\chi^2_{\rm red}$ 15.0–28.7 with two tuned parameters per galaxy, versus parameter-free MOND at 6.5). The highest scorer anywhere in the matrix is AeST at 5/8 — and AeST is dead, on precisely the gate the spec names as the killer (Cassini). The calibration cuts both ways and is the matrix's integrity check: a hostile published theory *can* score most of the items, so the spec is not gerrymandered to flatter the hybrid; and a high count with the wrong NO is still a dead candidate, so the matrix is gated, not additive. The sharpest output: **items 1 and 3 — the two ingredients the swarm *derived* (the inertia-side lever; the ultralight knee field in $[1.3\times10^{-29},\ 1.6\times10^{-24}]$ eV) — are universal NOs: no published field-level candidate, live or dead, has ever built either.** The hybrid must be built, not adopted. The honest converse is recorded at full weight: the superfluid family's phase architecture overlaps the spec more than any pre-2015 host ever did, its decisive gates (a computed Q_2 for the phase screen; a lensing-RAR break at the phase radius) are named, bounded, and uncomputed, and if those gates were run and passed, "build" would weaken toward "extend."

What the literature does supply is adoption-grade parts, banked alongside the spec. Three templates: the **phase/decoherence solar screen** (an existence proof that acceleration-keyed galactic dynamics plus solar safety is achievable without frequency dressing); the $\Phi = \Psi$ **lensing partner built from the medium's own 4-velocity** with no extra vector field (arXiv:1602.05961 §6 — the only such construction in print, owing a post-GW170817 c_T audit); and the **finite-temperature two-fluid machinery** for a phase-keyed type split (arXiv:1809.08286). And four imported kill-tests the falsification registry did not previously carry, which any built hybrid faces on day one alongside the banked eight: the GW170817 c_T audit, MOND-regime hyperbolicity (arXiv:2105.02241), stellar Cherenkov losses into any slow matter-coupled mode (arXiv:2103.16954, 2208.14308), and the Milky Way vertical-dispersion Jeans test (arXiv:1812.08169, 1911.12365).

6.4 The first construction attempt, and its pre-assembly kill

The build was then attempted — and was gated out before assembly, by the pre-registered fraction–amplitude calculation Section 6.1 named decisive: can an ultralight carrier limited to the structure-formation ledger (f $\lesssim$ 1–3% of the dark budget across the knee band) supply the *full* MOND amplitude, $g_{\rm obs} = \sqrt{g_{\rm bar} \cdot a_0}$? The answer is no — by every route, in every decade of the band, under every convention checked (a_0 footing, local density, ledger edges, charity dials: none moves any gap by more than ∼1 dex against shortfalls of 4.2–38 dex).

- **The sourcing route** (the superfluid template: the carrier gravitationally supplies the phantom mass; Milky Way pinned at $M_{200} = 10^{12}\,M_\odot$, NFW $c_{200} = 10$). Required carrier fractions: f = 0.85 at 50 kpc, 1.58 at 300 kpc, **3.27 at 1 Mpc** for a clustered carrier in the bound decades — i.e. even at f = 1 the measured Mpc-scale lensing amplitude needs ∼3.3× more mass than the entire pinned halo carries there (the published SfDM lensing kill, arXiv:2303.08560, re-derived as a mass budget). Against the ledger the clustered carrier is ×28–99 short inside R_{200} and ×109–327 short at 1 Mpc. In the homogeneous decades — everything at and below $\sim 10^{-26}$ eV, including the DR4 discriminator mass, where the field cannot cluster on a galaxy — $f_{\rm req}$ runs from 46 (1 Mpc) to $1.7{\times}10^4$ (50 kpc): gaps of $\geq 9{\times}10^2$ at the best radius and $\times 3{\times}10^5$–$3{\times}10^7$ where rotation curves live.

- **The mediator route** (the spec's actual structure: the carrier's condensate amplitude dresses the inertia channel). The route's own scaling is real and was exploited to its limit: $\varepsilon = \beta\sqrt{2f \cdot \rho_{\rm DM}}/(m \cdot M_{\rm Pl}) \propto \beta\sqrt{f}/m$ improves as 1/m, moving N3's coupling wall from $\beta_{\rm req} \sim 10^{40}$ down to $\sim 10^4$ — a 36-order improvement, the largest single step any bath-side channel has taken. **It is still not enough.** The best decade in the whole band (the floor, $1.3{\times}10^{-29}$ eV) reaches $\varepsilon = 1.1{\times}10^{-7}$ at the Cassini-allowed coupling — 7.0 dex short of the order-unity floor; at maximum charity ($\beta = 1$ with all fifth-force bounds ignored, the contested f = 0.1 window, $\rho_{\rm local} = 0.4$ GeV/cm^3, a ×100 soliton overdensity extrapolated beyond its validity) the gap is still **4.2 dex**; at the DR4 discriminator mass it is 8.9 dex (6.4 at $\beta = 1$). No decade passes.

- **The structural contradiction** (new, and convention-free): the MOND sign requires $m^2 < 2H^2$ (the N1/N3 tail channel), while the knee window forces $m \gg H$ ($m/H_\Lambda \in [1.1{\times}10^4, 1.3{\times}10^9]$) — **the sign and the knee cannot come from the same field's vacuum tail** (that route is in any case ≥38 orders over Cassini). The spec sheet is thereby sharpened: it requires two distinct functions even *within* the carrier sector.

- **Every exit lands on an already-banked kill:** quadratic couplings make the knee a local-baryon-density dial, re-creating exactly the environment-keyed RAR-scatter structure N5 killed at 4.4–5.2σ; derivative/P(X) couplings exit at the meV scale into the superfluid family's own mapped walls; a metric-MOND amplitude partner re-imports the Cassini Q_2 kill that took AeST and DEW; a lens-only slip sector (required gravitational slip 6–61× across the deep lensing bins) has no published realization and is squeezed by GW170817; and the dS bath itself is the N3 wall (10^{85}).

The verdict, both ways at full weight. The build as specced — a fraction-limited carrier supplying the MOND amplitude — **dies pre-assembly.** The framework-favorable findings ride along: the 1/m amplitude scaling is genuine (the door closed from ∼85 orders of impossibility to ∼4–7 — and is still closed); the sign–knee incompatibility is a clean structural discovery; and the Cherenkov gate genuinely passes at allowed couplings (recorded as a pass, with the honest reason: the coupling is too small to do anything — any coupling large enough to do MOND re-arms the gate). What the kill does *not* touch is the carrier as pure knee-setter with the amplitude paid elsewhere — but every published "elsewhere" lands on a banked kill, so as of this writing the specification has no surviving realization. The named reopen conditions: a worldline coupling with no static-limit fifth force whose in-band amplitude is not paid from the carrier's energy density (none exists in print); the fraction ledger collapsing by ≥ 4 orders (CMB, BOSS, eROSITA and Lyman-α would all have to be wrong together); or abandoning the order-unity amplitude — i.e. abandoning MOND as the target, which is outside the spec. One registry consequence is stated now and again at the predictions table: the Gaia DR4 wide-binary fork remains a real measurement, but it no longer discriminates *for* this build — the build it was to discriminate for does not survive to assembly.

6.5 The gauntlet: complementary failure

The banked kill-tests were then run numerically on the most-published object in the hybrid class — Berezhiani–Khoury superfluid dark matter, at its own published fiducials, with its authors' own escape reading carried at full weight alongside face value, under a pre-registration locked before any run. The result is a **complementary failure: the superfluid class clears exactly the two walls that kill universal-force MOND, and dies on exactly the fronts where universal-force MOND survives.**

- **Clears — and these are the program's two hardest lensing exposures.** The 40.5σ metric-passive wall, by construction (lensing reads real condensate-plus-halo mass, not baryons alone). And the $\sim 9\sigma$ type split: the predicted early$-$late lensing-RAR offset is positive (early above late) in all SHMR configurations and all 15 bins, the measured +0.261 dex sits inside the predicted bracket [+0.119 (type-blind SHMR), +0.401 (type-dependent)], and the phase chain is quantified end-to-end (68% of early-type lens halos above the condensation boundary versus 27% of late; median $\log M_h$ 12.36 vs 11.86). The locked wording is enforced: this is *consistency* for the real-lensing-mass class — the split that is fatal to type-blind universal laws is expected here — it is **not** "SfDM confirmed" (Brouwer's own MICE/BAHAMAS comparison simulations disagree about the split), and the same model still fails the *absolute* lensing-RAR shape on the same data in print ($\chi^2_{\rm red}$ 15.0–28.7 vs parameter-free MOND 6.5, arXiv:2303.08560).

- **Qualified by the mass-binned data (new since the gauntlet).** The *sharp* form of the phase mechanism — a condensation staircase in halo mass, demanding a bin4$-$bin1 offset of +0.344 dex (slope +0.297 dex/dex) on Brouwer's released mass-binned RAR profiles — is **refuted at** $\approx 7.3\sigma$ (measured bin4$-$bin1 $= +0.053 \pm 0.040$; slope $+0.041 \pm 0.037$, full released covariance). Both directions at full weight: a mild mass trend *does* survive at 2.0σ ($+0.122 \pm 0.062$ dex/dex) in exactly the 1-halo-safe bins, while the 2-halo-dominated complement is dead flat ($+0.000 \pm 0.047$) and the modeled 2-halo-induced slope is only +0.008 — the control inverts the mundane reading, which would put the trend in the dirty bins rather than the clean ones. The sign-match above therefore stands as a sign/type statement, but any phase-driven account of the split must make the condensation fraction a much smoother function of halo mass than the boundary model, or key it to a second variable (merger history, temperature) rather than mass alone.

- **Dies — face value, in the solar system, by 6–8 orders.** The published parameters put the Sun *inside* the coherent phase (thermalization radius 83–158 kpc), and no computed criterion shuts the phonon force off at planetary radii: the Sun is subsonic (Mach 0.39–0.55 — the named supersonic-disruption screen never engages), the disruption bubble is 0.005–0.01 AU, and the only real breakdown (superluminal phonon dispersion inside $\sim$10 AU) removes predictivity, not the force. Face value: perihelion drifts 5.9×10^6–$8.6\times10^7\times$ over ephemeris bounds; **Cassini $Q_2 = +4.50\times10^{-26}$ s^{-2} (24.1σ over the 2026 bound, BFK fiducial) to $+7.24\times10^{-26}$ (39.3σ, B-K 2015)** — worse than AeST and DEW, because the phonon force has no Newtonian-restoration knee at all — with a clean analytic byproduct of the run: the scale-free phonon interpolation gives $q_{\rm ph} = -3/7$ **exactly, independent of the external field** (verified to six digits across $e_N = 0.5$–4.0); and the solar reflex lands $\times5.4\times10^4$–1.8×10^6 over the agentE budget on the same Jupiter-synodic carrier. The authors' screened reading ("the phonon EFT breaks down near stars" — one published sentence, no criterion) escapes all of this **only by incompleteness**: no prediction inside $\sim$10 AU, where AeST and DEW made fatal ones. Both published fiducials are independently Cherenkov-excluded in print; the two-field fix evades Cherenkov but keeps the static force, so the face-value kills transfer to it intact.

- **SPARC, both ways:** at $\Upsilon = 0.5$ the phonon-only skeleton is just +0.008 dex behind the McGaugh baseline unweighted (and ahead of it weighted) — not SPARC-dead as a fit; at $\Upsilon = 0.70$ it is +0.034 behind, and both readings share a Υ-independent high-acceleration overshoot (+0.07–0.21 dex mean excess at y > 10, exactly where the data sit on the 1:1 line). The model does not predict a *single* RAR at all (the condensate term is galaxy-dependent), and the published 169-galaxy verdict is "unsatisfactory" (arXiv:2201.07282).
- **The wide-binary fork gains a third branch:** face-value SfDM predicts a **+26–37% deep-bin velocity boost** — the largest signal of any candidate on the board, opposite F4's $\sim$+4% — so Gaia DR4 adjudicates three ways in one shot (a clean $\sim$3% null kills face-value SfDM decisively; a +10–15% detection that confirms soft MOND shapes *also* kills it; only +25–35% selects it).
- **For the spec:** the B-K object is **not** the missing object. Its particle mass (0.6–1 eV) sits 24–29 decades above the knee band and cannot be moved into it (the phonon sound speed goes superluminal already at m $\approx 1.7\times10^{-3}$ eV; the thermalization radius overshoots the horizon by 23 decades at the band top); its knee $\bar{a} = \alpha^3\Lambda^2/M_{\rm Pl}$ is set by the symmetry-breaking scale, not by m, so the DR4 knee discriminator cannot probe it; and its a_0 is a fitted phonon coupling with no $\Lambda_{\rm cosmological}$ tie. The two hybrid classes are structurally disjoint objects that share a word. What the gauntlet *does* deliver is the first worked existence proof of the spec's lensing-partner phenomenology: a hybrid whose partner carries real, type-dependent mass demonstrably absorbs both of the program's lensing exposures — attached, in this realization, to a microphysics that fails everywhere else.

7 The derivation chain

The second edition's centerpiece: the program's results — Sections 3–6 plus the nine verdicts banked after the first edition — consolidated into one chain from Λ to the named effective law. The final pass folds in five further verdicts (agents X, AA, BB, Y, and the agent Z dial measurement): Links 5–7 below carry their final forms, and 7.6 carries the assembled action. Status legend: EXACT (machine-verified theorem/identity) · DATA (empirically certain, convention-audited) · CONDITIONAL (a published chain given named postulates) · NAMED (a published template) · CONTESTED · CLOSED-NEGATIVE (a proven boundary). The chain is continuous and broken in exactly three places; each break is bounded by a theorem or a live contest, not by ignorance. (Memos: `reviews/toe_law/DERIVATION_CHAIN.md` *plus the §7 block of the memo index.)*

7.1 Links 0–3: the exact substrate, and the theorem that closes the thermodynamic lane

Link 0 — Λ exists and is measured (DATA), footed on ρ_{DE} (Section 2.1) — a footing independently converged upon from outside: Verlinde's own closing caveat (arXiv:1611.02269 §8: a_0 "should actually be defined in terms of the dark energy density") and the quasi-de-Sitter emergent-gravity literature (arXiv:1612.06282, 2206.11685), which drifted to a_0 footed on $5.41\times10^{-10}\ \mathrm{m\,s^{-2}}$ — numerically cH_Λ — because the data pushed it there. **Link 1 — Λ sets a temperature (EXACT):** $T_{\mathrm{dS}} = \hbar H_\Lambda/2\pi k_B$ (Gibbons–Hawking).

Link 2 — accelerated matter in dS sees $T_{\mathrm{eff}} = (\hbar/2\pi c k_B)\sqrt{a^2+(cH)^2}$ (EXACT), plus a new theorem. Deser–Levin; N1's closed-form commutator (Section 4.4) extends it. The new theorem (agent Q, 26 sympy checks): **the Deser–Levin temperature is identically the Tolman-shifted Gibbons–Hawking temperature** — $\sqrt{a^2+H^2}\cdot|\xi| = H$ exactly for the static dS observer. The consequence closes a lane exactly: Jacobson's Clausius construction of the Einstein equation (arXiv:gr-qc/9504004), rerun with T_{DL}, changes nothing — the $\sqrt{a^2+H^2}$ cancels against the Tolman weight of the locally measured heat identically, at all orders in H/a. Jacobson-with-Deser–Levin = Jacobson; the H^2 term propagates nowhere; Λ stays the Bianchi integration constant (the null balance is provably Λ-blind). In the only readings where H survives (Tolman-broken bookkeeping) the correction is exactly $\mu_{\mathrm{F4}} = a/\sqrt{a^2+H^2}$ on the *gravity* side — anti-MOND, with total shutoff below $g_N = cH$, the sign quadrature-forced. The finite-$(H/a)^2$ coefficient is pure scheme ($\{0, 1/6, 1/3, 1/2\}$); a companion theorem (agent T) shows the Guedens–Jacobson–Sarkar $O(x^2\cdot\text{Riemann})$ Killing failure **vanishes identically in exact dS** (Killing fields vanishing at a point are exactly linear in Riemann normal coordinates — verified three ways), so the identity holds unqualified; and at a = a_0 — $(H/a)^2 = Z^2 \approx 33.5$, 89% of the static-patch depth, the flat wedge overshooting the horizon by 314% — the construction ceases to be *local* and becomes Gibbons–Hawking thermodynamics of the cosmological horizon itself, the Λ sector already in the Einstein equation. **The closure in one exact sentence: Clausius consumes T, which is κ-only and Tolman-trivial; MOND needs dT/da, which lives in the dissipation kernel the equation of state never touches — $\delta Q = T\delta S$ provably cannot do it.**

Link 3 — the bath knows *where* a_0 is (EXACT), and the μ_{F4} triple echo. $\kappa = \sqrt{a^2+(cH)^2}$ bends at cH: the length scale is not inserted; it appears. Alongside, a structural echo at full weight: $\mu_{\mathrm{F4}} = a/\kappa$ has emerged spontaneously in **three independent places** — the bath susceptibility that selected F4 ($m_{\mathrm{eff}} \propto dT_{\mathrm{eff}}/da$, Section 4.2); agent Q's Clausius balance (exact identities $a/\kappa = T_{\mathrm{Unruh}}/T_{\mathrm{DL}} = 2\pi\cdot dT_{\mathrm{DL}}/da$); the dimensionless static-patch radius of the stationary worldline family, $Hr(a) = a/\kappa$ exactly (agent T). **Recorded as a structural echo of the quadrature algebra, not a derivation**: nothing in the three appearances fixes the inertia-side orientation or the coefficient, and the second applies the kernel on the anti-MOND leg.

7.2 Link 4 — the coefficient: a two-candidate contest, geometry-audited (DATA + CONDITIONAL)

The framework's $Z = \sqrt{32\pi/3} = 5.789$ is **data-certain and underived** (Section 2.2). The only derivation-flavored rival (agent P): Verlinde's $a_M = a_0/6$ (arXiv:1611.02269, eqs. 7.40/7.43) is **dimension-forced — $(d-3)/((d-2)(d-1))$ at $d = 4$, zero tunable freedom, zero data input — given five named postulates** (volume-law dS entropy; linear-in-r interpolation; the elastic dictionary; equality-saturation of an inequality his own §7.1 stops at; spherical/static/isolated equilibrium). The condition carries full weight: if the postulates fail (the unresolved Dai–Stojkovic dispute), there is no MOND term and no coefficient. But within the candidate chain the 6 is forced where Z is selected — opposite epistemic signs. Empirically the two are degenerate and will remain so: splitting the 3.65% gap at 3σ requires $\sigma(a_0) < 1.2\%$,

$\sim 20\times$ below the honest systematic floor; on the program's own 175-galaxy SPARC pipeline the Υ-profiled scatter difference between $cH_\Lambda/6$ and cH_Λ/Z is 0.0002–0.0012 dex — pure noise; and they are **mutually exclusive as exact claims** (6 rational; $Z^2 = 32\pi/3$ π-bearing). The geometry route was then run pre-registered (agent T): does the Jacobson-construction breakdown acceleration carry a forced coefficient? **O(1)-NULL** — no symbolic match to $1/\sqrt{32\pi/3}$ under any of three breakdown definitions (one has no finite-a solution; one is identically zero in exact dS; one never fires); every finite candidate is threshold-set, spread $\times 39$ — the spread is the answer; near-misses flagged as bait where they arose (the merger misfit at a_0 sits 0.19% from π: transcendental coincidence). **The break is bounded: either Z stays data-selected, or Verlinde-style conditional counting is right and the constant is 6 — and no RAR-class measurement tells them apart.**

7.3 Link 5 — the mechanism: CLOSED-NEGATIVE at the worldline; final form — one object, one property, at theorem + data grade

The first edition closed the worldline lane for point detectors at all orders (Section 4.3.3) and bath-coupled light scalars (Section 4.4); Link 2 closes the thermodynamic lane by theorem (any mechanism must couple to dT_{eff}/da on the inertia side). The last opening — extended detectors — now closes too, **with its physics confirmed (agent L)**. The premise is true: the dS bath's coherence length is the Hubble radius (exact two-worldline invariant $|\Delta P|/P = (Hd/c)^2/2$), the monopole charge couples whole, $\varepsilon_{\text{body}} = N \cdot \varepsilon_1$ exactly, and internal stellar temperature costs only $(v/c)^2/2 \approx 2 \times 10^{-6}$. The triad then kills it three independent ways, each alone sufficient: (a) *universality* — $\varepsilon \propto M$ predicts a 0.55–1.0 dex gas-versus-star RAR split against one tracer-blind relation at 0.057–0.13 dex, lunar laser ranging adding $\times 5$–2×10^7; (b) *the solar reflex is strictly worse* — the Sun is the maximal-N body and the kill is 100% Sun-carried: $\times 2\times 10^5$–8.6×10^7 over budget; (c) *magnitude*, on an exact identity — $\varepsilon = GMH/c^3 = r_g/2R_H$: coherent dressing saturates at Schwarzschild-over-Hubble, so MOND strength requires a horizon-mass body (a star is 28.3 dex short; a $10^7\,M_\odot$ cloud, the largest common-acceleration unit, 21.3 dex; the horizon baryon budget 6.9 dex). One line: **the N^2 gain and the WEP violation are the same number.** The closure now spans point, composite-point, and extended detectors.

The final form (agents V + X + AA + BB, convergent): the mechanism is an active, dS-invariance-breaking medium — the khronon sector itself, pumped by the Λ/dS energy budget — and exactly one property of it remains underived. Three results lock it. (i) *The kernel inversion (agent V): KERNEL-EXISTS-BUT-ILLEGAL.* The inversion is unique and explicit; the required spectral density σ_{req} is computed (a fourth-root essential singularity at the lightcone, all inverse moments zero); de Sitter Källén–Lehmann positivity closes every invariant carrier (α-vacua, interacting fields, squeezed states) — the surviving carrier must break de Sitter invariance, i.e. precisely the khronon medium of Link 6's construction, which inherits σ_{req} as its derivation target. (ii) *The causality theorem (Theorem X2, stated in full in Section 7.4):* **no passive vacuum closes any causal MI** — the secular channel is irreducibly active (the medium's mandatory co-payment, $(1/\mu - 1)\times$ the external power, measured at $\times 2.58$ in the self-consistent run), and the invoice names the Λ/dS bath as the only reservoir in the action with the throughput to pay it. (iii) *The escape closures, at data grade.* The sign question is closed two ways: agent AA's fixed-convention reconciliation (26/26 machine checks, anchored on both sides by the dS mass-loss runaway and the flat-space scalar-Yukawa *excess*) confirms the banked assignment — **the inertia-DEFICIT channel is $M^2 \equiv m^2 + 12\xi H^2 < 2H^2$, intrinsically de Sitter** (flat space gives an excess; V's contrary flag traced to a vector-signed Yukawa anchor on a force-only object, and retired) — and the SPARC inversion independently selects the same assignment as the only MOND-capable one; the banked sign–knee disjointness (Section 6.4) stands at full strength. And the one legal escape the inversion theorem left — a dS-invariant, Källén–Lehmann-positive, conformal-balanced spectral mixture — is **ESCAPE-CLOSED on both pre-registered arms** (agent BB): the best legal mixture lands +0.026 (framework) / +0.023 (canonical) dex above the locked SPARC baseline — past the pre-registered escape-closure line (+0.020) at both footings — and every mixture that even approaches the band fails the solar-reflex budget $\times 20.8$ through a forced-positive variance pincer (conformal balance makes the solar observable a positive-definite second moment: the band fit needs spectral variance ≈ 5.0 where the Sun allows ≤ 0.24 — no sweet spot exists), with the failure mode exactly V's predicted fingerprint (a deep-band plateau plus a cubic-log tail). The boundary theorem is therefore theorem + data grade: **no linear field-bath in dS-invariant QFT — legal, tuned, or even sign-illegal within the same kernel basis — carries the rotation curves.** What survives of Section 6's specification is one named object with one underived property: the khronon medium and its $\sigma_{\rm req}$. Corollary, a new falsifiable prediction (Section 8, row 9): no kernel of any kind produces the $\sqrt{a/a_0}$ deep-MOND onset — μ must flatten to constant + $O(a^2)$ below some acceleration $a_\star$; for any surviving mechanism $a_\star$ sits below x $\approx$ 0.05, since the legal family died precisely by its flattening intruding into the SPARC band. *First confrontation (agent CC, addendum): existing deep kinematics prefer $a_\star = 0$ everywhere — SPARC's deep decade (1,179 points to $y = 2.5 \times 10^{-3}$) at both footings, both baseline shapes, and 87% of galaxy bootstraps — bounding $a_\star \leq 0.107\, a_0$ (95%), a factor $\sim$2 weaker than the band ceiling: the prediction is alive and untested in its allowed window. The only published deep downturn is environment-keyed (EFE-shaped), not acceleration-keyed (floor-shaped); the decisive discriminator is registered — isolated ultra-deep rotators, where a floor turns down at one $g_{\rm obs}$ everywhere and the external-field effect does not act.* **The $\sigma_{\rm req}$ scoping (agent EE, addendum): STRUCTURALLY-CAPABLE — for the pumped khronon; the minimal khronon cannot, by machine theorem; Link 5 becomes a defined calculation.** The positivity no-go is evaded outright, not negotiated: the khronon admits *no* de Sitter Källén–Lehmann representation for any measure class (an O(1) irreducible residual, machine-quantified), and the replacement two-variable spectral representation over the residual symmetry group is derived — the no-go's load-bearing step has no analog at the worldline, even though stationarity and thermality survive family-wide. An exact theorem rides along: the broken group's dilatation orbits ARE the Deser–Levin trajectory family ($a = b\kappa$ with $\kappa^2 = a^2 + H^2$, machine-exact, anchored on the banked conformal kernel) — the khronon's residual symmetry natively contains the accelerated-detector family the framework is built on. The free khronon cannot carry the kernel (its lightcone cut tail is identically zero — an absent object, not a wrong exponent), and a Bogoliubov lemma bars occupations and squeezing from the dissipation channel: **the pump must modify the dynamics** — Theorem X2 re-derived from the worldline side, the third independent convergence on the active medium. The required pump fingerprint is computed and saddle-verified ($\Delta\tilde{\rho}_c(\omega) \sim A\,\omega^{-1/3} e^{-\tilde{c}\omega^{1/3}} \cos(\sqrt{3}\,\tilde{c}\,\omega^{1/3} + \tilde{\varphi})$, $\tilde{c}$ computed per footing, positivity window $A_{\max} \approx 5.7$), with matching conditions (C1)–(C5) written: what remains of Link 5 is one calculation — the scale-invariant gain profile $g(k_{\rm phys}/H)$ of the Λ-pumped khronon, checked against them. The hostile negative is kept at full weight: a^2-analyticity at $a = 0$ extends to the whole scale-invariant khronon class — the deep-MOND flattening floor $a_\star$ is now this route's *own* prediction, making the registered discriminator decisive in both directions.

7.4 Link 6 — the effective law: NAMED; the covariant home BUILT at the EOM level, conditional on Link 5's reservoir

The law (agent M). Milgrom's 2022 time-nonlocal modified inertia (arXiv:2208.07073) — $m\,\hat{a}(\omega) \cdot \mu[\mathcal{A}(\omega)/a_0] = \hat{F}(\omega)$, $\mathcal{A}$ a frequency-filtered functional of the whole trajectory, $\theta(1) = 1$ anchoring rotation curves — was run through the full nonrelativistic battery. The structural finding: the filter is inert against the solar reflex ($\theta(1) = 1$ pins the Sun's own-frequency response; suppression delivered 0.72–0.78 versus required $\leq$ 0.072–0.117), so the agent-E kill transfers to the power-law-μ members (standard μ $\times$6.1–10.9 over budget at the physical footings, $\times$205–222 hostile; simple μ $\times 3{\times}10^4$) — **while the exponential-tail (McGaugh-RAR) member**

passes by more than 10^{13}, with precession relaxed ($\times$3.4–6.8, sign-flipped) and SPARC at the acceleration-keyed baseline (0.1950 dex). **Milgrom-2022 + exponential μ-tail is the first published object to clear every nonrelativistic matter-sector wall the program has built** — WEP-safe, conservation-complete nonrelativistically, composite-body-solved, p $\equiv$ 0 acceleration-keyed — and is adopted as the matter-sector template. The adoption sharpens Link 5: the mechanism must produce specifically the exponential tail; the filter contributes nothing to the selection — the solar system performs it. It also reshapes the wide-binary fork (Section 8, row 1): $\theta(0)$-enhanced EFE suppresses the soft-shape boost $\times$2–4, and an intermediate +4–8% DR4 amplitude would positively select M22-style enhanced-EFE modified inertia.

The covariant home (agent U): BUILDABLE. A khronon-framed lift, constructed on paper and run through the relativistic gates with maximum hostility: gravity pure Einstein; a canonical khronon supplying the unit timelike frame u^{μ}; the M22 functional evaluated in the u-frame. Verified unbuilt first — INSPIRE's full citing list of arXiv:2208.07073 contains no covariant lift; every published khronon/aether-MOND is modified *gravity*, the class Section 3 killed. The lift buys exactly one thing: the frame problem — the structure the program's covariance memo proved no metric-local construction can supply — solved at a pinned, survivable PPN price (generic corner $\alpha \lesssim 8{\times}10^{-7}$; the tuned corner at the α_1 ceiling newly found to sit at the vacuum-Cherenkov edge). Cassini Q_2 is **absent by architecture, not screened** (no force sector to source it); WEP holds in three verified layers; the Bruneton singular surface does not transfer; agent M's battery transfers at the <1% level — and, against the framework's interest, the lift does *not* manufacture the N5 corridor dressing (covariant M22 stays at p = 0). The one open gate it left — conservation/causality of the retarded worldline functional, the historical death-spot of covariant modified inertia, with Schwinger–Keldysh (arXiv:1712.07066-class) the named route — has now been run. The minimal cosmological writing makes a_0 a constant coupling $\equiv$ the framework's pure-Λ $\sqrt{\rho_{\rm DE}}$ branch; the khronon's one natural local scalar, $\nabla \cdot u$, writes the rival rising $a_0 \propto H(z)$ branch: the a_0(z) fork becomes a property of one term in the action, adjudicated by the data program.

The gate run (agent X): BUILT at the EOM level — PARTIAL, with the obstruction converted into a theorem. The Galley/Schwinger–Keldysh doubling delivers a strictly retarded, u-clocked, spectrum-resolving (adaptive-window) writing of the M22 functional that reproduces agent M's banked inventory on quasi-stationary worldlines ($\mathcal{A}$-level drift 0.02–0.03%; the exponential-tail solar reflex to print precision), has zero pre-acceleration where the symmetric form pre-responds at 64% of the full quiet$\rightarrow$loud shift, no runaways (monotone $x\mu$; damped Picard $\leq$ 34 iterations), and closes total $\nabla_{\mu}T^{\mu\nu} = 0$ on-shell by the doubled-action Noether identity with the khronon as the energy reservoir — the ledger verified at 10^{-14}. Two structural findings ride along: the khronon is forced *twice* (filter frame and causal clock — a covariant retarded memory needs an invariant "how long ago"); and a *fixed* causal kernel is dead on arrival (the Kramers–Kronig-forced Im $\hat{\mu}$ pumps epicyclic perturbations at O(0.1–1) per radian in deep MOND — only the adaptive construction survives, residual flux $\propto N_{\rm cyc}^{-1}$). The PARTIAL: **Theorem X2** (the passivity sum rule) — causality plus vacuum passivity force $\hat{\mu}(0) \geq \hat{\mu}(\infty)$ on the linearized response, while the scale-invariant deep-MOND limit of *any* modified-inertia theory forces the inversion — so the conservation proof sources the khronon with an active-sign flux no vacuum can lawfully supply: the khronon is carrier and clock, *not battery* (its PPN-corner stockpile drains $\times$50–3600 short of a Hubble time), and the build is conditional on Link 5's pumped reservoir, whose invoice is priced — throughput $\sim 10^{33}$–10^{35} W per L_*-galaxy, covered by the Λ/dS bath with $\times 10^{2}$–10^{4} margin at the 100-kpc-box level (a conservative box: real L_*-galaxy domains are $\times$2000 larger) and $\sim$8 further orders at the per-galaxy-honest horizon level — the build memo's "$\sim$15" compared a single galaxy's bill to the global reservoir and is corrected here per the hostile audit. The flux is invisible everywhere precision data live ($\times$14–22 inside the hostile reflex budget; $\leq 10^{-48}$ at every planetary channel);

the construction's falsifiable novelty is a transient fingerprint — a freshly-kicked deep-MOND worldline responds with temporarily *enhanced* MOND behavior (×2.32 measured — the addendum confrontation below finds this reading-dependent) for $\sim N_{\rm cyc}$ cycles, invisible at high x. **Links 5 and 6 are thereby provably one problem**: causality alone, pushed through Schwinger–Keldysh, re-derives the dS-bath-class reservoir as mandatory, with sign and amount. The hostile reading rides at equal weight: the theorem legitimizes the dS-bath necessity only if one already accepts MOND-as-inertia — read cold, it says this class of theory cannot be closed by any conventional healthy field sector; the pump coupling is exactly as unwritten as Link 5's mechanism was before the gate ran; the adaptive window introduces a dimensionless choice ($N_{\rm cyc}$) nothing yet fixes; and nothing in the construction derives Z, a_0, or the exponential tail — it consumes all three. *Hostile audit (agent FF, addendum): X2 SURVIVES-STRENGTHENED.* The theorem is re-derived from scratch by a bath-microphysics route (every positive mode adds DC inertia; 2,000 random baths never invert the ordering) and **extended to nonlinear passive media** by a two-line storage-function argument — the MOND secular trajectory requires $(1/\mu - 1)\int F\,v\,dt > 0$ of free energy from a medium bounded below in its ground state, a contradiction; 30 adversarial nonlinear baths reach at best R = 0.30 where MOND needs 3.58, and only an energy-unbounded ghost sector fakes it (the option already excluded). The non-L1 kernel tail (a real $1/\tau^2$ tail, verified to 0.3%) does not break the Fourier-limit statements (independent adaptive-quadrature Hilbert transform matches to 0.02–0.17%). The EOM survives step and impulsive-pulse attacks with machine-zero pre-acceleration; the per-event transient co-payment is geometry-specific (×7–34 on quiet worldlines, not the build memo's ∼1% — the ×242 margin still clears it). The 0.03% agreement is reclassified as a shared-source regression test, and the solar anchor is separately re-confirmed to <0.2% through a fully independent constants chain (DE440 system GMs, JPL J2000 elements, GRAVITY-era galactic line, Planck-2018 footings re-derived). And the question the build never asked signed — which way the energy actually flows — comes out the right way: the reservoir drains, secularly (span ×4 → loss ×4.64), at 2.63× the external work in deep MOND: the X2 invoice direction, exhibited independently by the dynamics. *The transient fingerprint confronted (agent JJ, addendum) — and an operator-form split found.* The banked ×2.32 turns out to be a prescribed-μ statement, not an action-EOM result: the construction carries two operator readings of its central relation that disagree on ALL transients — the implicit-μ reading reproduces the ×2.32 enhancement, while the literal action EOM gives 0.70× *suppression* on the same step (a quasi-Newtonian floor lasting $\sim 2.6\, N_{\rm cyc}/\langle d(x\mu)/dx\rangle$ orbits). The fingerprint's sign is an unfixed operator-form convention alongside $N_{\rm cyc}$. First data confrontation: six published tidal dwarf galaxies (0.2–0.8 internal orbits old) sit $-0.63\ldots-0.68\pm0.09$ dex *below* the isolated RAR — the naive "+0.37 dex above" transplant dies at ∼11σ — while the construction's own EFE-embedded prediction $(+0.02\ldots+0.07$ dex: the parent's external field erases the enhancement) passes against the settled-EFE benchmark (-0.13 ± 0.10). Non-diagnostic between the two operator readings on this sample (0.1–3.1σ across convention forks); the decisive test is registered — isolated young deep-MOND rotators split the three readings at ≥0.3 dex; one clean object decides (watch entry 13).

7.5 Link 7 — the lensing partner: necessary (first edition), UNIQUE as a class — and its scalar carrier now closed constructively

The double-counting theorem (agent W, Part 1). The first edition proved the partner necessary (40.5σ, Section 5.2); what had been asserted but never computed was the other jaw — a real-mass partner supplying the lensing-demanded phantom profile pulls stars too. Now a theorem, with an exact lemma at its spine: for monotone $x\mu(x)$ (Milgrom-2022's own uniqueness condition) the modified-inertia response to a total real field is unique, so rotation curves satisfying the observed RAR force $g_N = g_{\rm bar}$ — **the real partner density is forced to zero everywhere kinematics probes, while lensing demands it nonzero at** 40.5σ**;**

the class cannot have both. Computed on the program's own SPARC pipeline: deep-RAR overshoot 0.165–0.49 dex at 8.7–21.6σ under every convention combination (the weakest cell, disk M/L charitably slammed to its grid floor, is 8.7σ; the honesty row — a statistics-starved deeper cut at 4.7σ with a *growing* offset — is reported, not hidden). The escapes close: the "partial cancellation" ordering is *worse* (unbounded $y^{-1/4}$ overshoot; 14.2–21.6σ); freeing the MI scale drives $a_{0,\mathrm{MI}} \to 0$ — particle dark matter with the modified inertia deleted; no partner fraction opens a window (joint tension never below 12.0σ). **Real-mass lensing partners are excluded as a class for modified-inertia dynamics.**

The unique survivor, scoped (Part 2): a metric-level, Ψ-channel (lens-only) slip sector — $(\mu, \Sigma) = (1, \nu)$: photons see the MOND amplitude, matter sees Newton, so the solar-system battery that killed AeST, DEW and face-value superfluid dark matter evaporates by construction. The narrowing is forced: photon-sector realizations are executed by GW170817's *differential-Shapiro* test ($\sim$1000 days of photon–GW delay difference versus the observed 1.7 s; Boran, Desai, Kahya & Woodard, arXiv:1710.06168); real-stress sources are closed by the causal-medium bound $|\pi| \lesssim \rho + p$, re-importing the double-counting overshoot $\times \sim 2$; what survives is slip in the metric itself, on the c_T-preserving beyond-Horndeski/DHOST branch, priced by the Saltas-class theorem (arXiv:1406.7139) in tensor-sector modification — the falsifiable handle. Passes and gates both named: solar slip clears Cassini γ by $\times 1.3\times 10^7$ unscreened; clusters re-fail $\times 1.97$ by construction; the type split forces a second variable, smooth in halo mass (the sharp staircase refuted at $\approx 7.3\sigma$, Section 6.5; a smooth 2.0σ trend survives with the control inverted); a naively ν-keyed slip inherits the absolute lensing-RAR shape tension (8.6–12.5σ). **No published realization exists** — the named near-misses each disqualified for a stated reason.

The scalar carrier closed (agent Y): OBSTRUCTED, constructively. The explicit build was attempted — a slip scalar on the assembly's own u-frame, the full DHOST-class operator basis at phantom order — and four machine-derived walls close every routing. (1) First-derivative operators cannot slip: the rr-constraint forces $\Psi' = \Phi'$ exactly (a first-derivative u-coupled scalar is modified *gravity* — the Cassini-killed class). (2) Two-derivative bilinears break c_T *in halos* by $\sim$15 orders at slip-matched amplitude — a new boundary with reach beyond this program: the published $c_T = 1$ DHOST classifications protect only timelike (cosmological) backgrounds, so any "DHOST slip at galactic scales" proposal hits the same wall. (3) The unique c_T-safe generators — the mixed cross-quartics — *can* track all four banked ν shapes exactly (closed-form, 10^{-15}) in the lensing channel, but at that amplitude feed the Hamiltonian constraint at 10^7–10^8 $\times$ the phantom, independent of kinetic stiffness across two decades — seven orders over the double-counting bar. (4) The exact-μ branch analysis collapses every remaining solution to trivial slip; the one elegant exception ($J = Y/4$) is the singular surface of the field equations, caught by its own consistency gate. **The kill is not ghosts** — the foliation makes every operator time-derivative-free in unitary gauge; the place the class was expected to die is exactly where it survives. What survives the obstruction is banked as architecture for the successor: $c_T \equiv 1$ and $\alpha_M \equiv 0$ *identically, on all backgrounds including halo interiors* (stronger than any published DHOST tuning); FRW quietness (the graviton-decay kill evaded by construction); and a constructive gem — the c_T-safe slip generators assemble into transverse-divergence (field-line-bending) operators: a built-in *morphology-tracking* amplitude at fixed g_{bar}, with the right sign (spheroids above disks) and sufficient range (the $\times 1.56$ needed is covered). That dial lands exactly on the program's own measurement (agent Z, on the 181k-lens re-measurement stack): the type split is **TYPE-IRREDUCIBLE** — at fixed stellar mass, redshift, *and* local density the early/late contrast stands at +0.194 dex (6.2σ combined; 1.5σ stratified, direction unanimous), and no measured dial (M_*, z, density) reaches the pre-registered $\geq 3\sigma$-at-fixed-strata bar — the second variable tracks morphology itself, exactly what the surviving generators supply. The class narrows once more: the partner, if it exists, is a **vector or spin-2 carrier on the same u, or a nonlocal/history operator** (the

M22-echo direction), with two named low-prior routes (the singular-surface exact treatment; the extended dressed bases) logged open, not closed. **The vector carrier closed (agent DD, addendum): SAME-WALLS — by the keying theorem.** Wall 1 is genuinely evaded (the condensate/hedgehog vector slips — a structural capability no scalar has) and Wall 2 by construction ($c_T \equiv 1$, $\alpha_M \equiv 0$ identically, machine-verified in-halo); but Wall 3 transfers, measured ($+2.3\times10^6 \ldots + 4.5\times10^7 \times g_{\rm bar}$ at slip-matched amplitude — opposite sign, $\times3$ smaller than the scalar's, equally dead at 5–7 orders over the double-counting bar), and Wall 4 transfers in closed form: the exact lens-only condition's geometric class is $\alpha^6\,(\text{slip}/\Phi') = 0$ — the pollution's irreducible core IS the slip. The root is not the carrier: **any local Y_a-keyed, MOND-amplitude slip carrier on the u-frame — scalar or vector, sourced or condensate — pollutes the matter channel at $(a_0 r/c^2)^{-1}\times$ phantom**, because the keying's own lapse response feeds the Hamiltonian constraint whatever field carries the operator. Banked residue: a zero-slip counterterm family (a matter-channel tool no scalar basis had); the GW170817 in-halo boundary extended to vector kinetic sectors; and the morphology dial proven *two-carrier-robust* — a design principle of the operator geometry, not of any particular field. **The nonlocal/history class closed (agent KK, addendum): OBSTRUCTED — by static equivalence.** The hoped-for escape — key the slip on the M22-style filtered acceleration history instead of the instantaneous acceleration — dies at the same wall: on a static background the filtered key's constraint response IS the window's DC weight, which tracking normalization forces to one; value and sensitivity are the same number, and the pollution reinstates at full amplitude. The theorem (static equivalence): any time-translation-invariant history key with a differentiable static read gives static field equations *identical* to the local theory — the keying theorem, the pollution tables, and the closures transfer verbatim to the entire time-nonlocal class (certified by reproducing the banked pollution row to machine zero through two nontrivial filtered reads). No frequency middle ground exists (10^{-7} suppression needs a window 111× the age of the universe — which never fills, so no slip; a tracking window keeps 72–99.6% of the pollution), and all three structured escapes fail: derivative-coupled keys read nothing static (zero on static fields and circular orbits alike); the counterterm family provably cannot reach the pollution's counterterm-free core; spectral keys read statics through $\theta(0)$. A convergence conjecture raised the same day — "u-frame nonlocality in both channels" — is computed false and retired: the matter sector's nonlocality is a dynamics demand, while the lensing job is static, where history is invisible. **Link 7's state after the round, stated plainly: scalar, local-vector, and history-keyed realizations are all machine-obstructed; the lensing exposure stands unexplained by every construction route explored to date.** The survivor space: the singular-surface exact route (low prior — nonlocal dressing provably cannot assist it), spatial nonlocality (unexplored, carrying no convergence argument), and non-b⊗b spin-2 condensates (keying-argument disfavored, not machine-closed). Both ways at full weight: the closed classes inherit every static success unchanged — exactly as capable, and exactly as dead, as each other.

7.6 The unified action: the blueprint assembled *(final pass)*

The first edition specified the missing object (Section 6); the second edition named it in both halves (7.4–7.5) under the locked wording "the blueprint is named and specified — not found, not built." The final pass assembles the halves into a single action scaffold with its construction slots resolved; the wording is locked before the equations: **this is a candidate-theory scaffold — named, specified, gated — not validated.**

$$S = S_{\rm EH}[g] + S_u[g,u] + S_m[\text{matter};\, g,u] + S_{\rm slip}[g,u]$$

$S_{\rm EH}$ is pure Einstein–Hilbert + Λ — gravity is *not* modified, which is what makes Cassini Q_2 absent by architecture — and the Λ here is the same Λ that sets a_0: one constant, two roles, the framework's distinctive content. S_u is the khronon sector supplying the unit time-like frame (PPN/Cherenkov corner), the structure Milgrom's theorem requires, no metric-local construction can supply, and causality forces a second time as the memory clock (7.4). S_m carries the M22 time-nonlocal functional in the u-frame with the exponential μ-tail, in agent X's causal doubled-variable writing. $S_{\rm slip}$ is the lens-only Ψ-channel sector coupled to the *same* u — the interface the theorems force, with the arXiv:1602.05961 §6 u^μ machinery, decoupled from its matter force, the named template — scalar realization closed (7.5), surviving carrier class named. The field equations follow by variation: δg gives Einstein + Λ with the three-sector stress structure forced by the banked theorems — $T^{(\rm slip)}$ contributes *only* anisotropic Ψ-channel stress at galactic scales (zero clustering stress-energy: the double-counting theorem), $T^{(u)}$ is bounded khronon stress ($\lesssim 2.5{\times}10^{-5}$ fractional), $T^{(m)}$ carries the inertia modification; δu gives the frame equation with both matter-side sources (the filter's frame-dependence, its PPN feedback exponentially zero at precision bodies; the slip coupling); δx gives the worldline law $m \cdot \mathbb{I}[a_u; a_0, \theta] \cdot a_\perp^\mu = F^\mu$, photons on null geodesics of g — lensing carried entirely by the slip-sourced Ψ channel. The consistency spine — total $\nabla^\mu T_{\mu\nu} = 0$, the nonlocal matter sector's energy deficit carried by the aether flux — is the assembly's deepest internal gate, and agent X's ledger closes it at the EOM level (10^{-14}).

The parameter inventory — the entire freedom of the assembled candidate: Λ (measured); $a_0 = c^2\sqrt{\Lambda/32\pi}$ (fixed by Λ; Z the one underived number); the filter θ ($\theta(1) = 1$ fixed by rotation-curve normalization; $\theta(0)$ bounded by the DR4 fork); the μ-tail (fixed to exponential by the solar system — no longer a choice); the khronon c_i (confined to the PPN/Cherenkov corner); the slip normalization (fixed by the lensing-RAR amplitude); and the dial (TYPE-IRREDUCIBLE per agent Z — morphology-tracking, now structurally located in the surviving field-line-bending generators rather than unnamed). The four construction slots are all resolved: **X** (the causal conserving EOM — built, conditional on the pumped reservoir), **V** (the mechanism fingerprint — $\sigma_{\rm req}$ computed, inherited by the khronon medium as its derivation target), **Y** (the slip Lagrangian — scalar obstructed, class narrowed, architecture banked), **Z** (the dial — type-irreducible, located in the operator geometry). What the assembly is, and is not: the first single action written to the program's full specification, every term forced or bounded by a banked theorem or measurement, with exactly one underived number — and not validated. The post-assembly gauntlet has begun. **The Boltzmann/CMB audit is run (agent II, addendum), and it kills the naive linear-scale extension convention-robustly** — the framework's own footing gives the *smallest* kill, so the deficit is real: linear structure is sub-a_0 everywhere below $z \approx 122$–155; the raw law over-lenses Planck's $\phi\phi$ amplitude by $\sim$1.7 orders (right direction for the A_L anomaly, wrong magnitude) and misses E_G by 73σ with a sign-flipped ISW; the computed EFE-regulated floor falls short $\times$112–777; and the only global cap that would fit cosmology ($\nu \leq 1.03$–1.14) sits *below the entire measured galactic range* ($\nu = 4.4$–306), so no scale-blind saturation can serve both. The slip sector therefore carries a named requirement: **a second discriminant beyond** $g_{\rm bar}$ (suppress $\Sigma-1$ by $\geq$50–800$\times$ at $k \leq 0.3$ h/Mpc while preserving halo ν to 1–3 Mpc) — currently unowned after the history-keyed candidate's same-day closure; the growth debt ($\mu = 1$ without CDM at linear scales) is logged as its own open. The cluster $\times$1.97 re-failure stands (the assembly does not fix it without the dial extending there); explicit-model verification of every inherited pass remains; and the TOE claim stays gated on Link 5's $\sigma_{\rm req}$ — now a defined calculation — and Link 8's quantum derivation.

7.7 Link 8 — the quantum gate: unmoved at the algebra, edge-wounded at the observables

The sweep (agent R). A 60-paper sweep with forward citations through 2026-06-10: GATE-UNMOVED — nothing published derives the DSSYK vacuum placement (the banked verdict stands: the deep-MOND sign of the de Sitter–microscopic entry is a 1:1 readout of an *assumed* spectral placement, center $\theta = \pi/2 \rightarrow$ MOND-favorable, edge $\rightarrow$ anti-MOND; the chord algebra supplies natural states of both kinds and cannot pick). The center camp added three consistency checks at an assumed placement (arXiv:2604.21014, 2506.02109, 2605.03037 — volume is sociology, not derivation); the edge camp is dormant (five citations, none adjudicating); a three-cornered reframing risk (center $\rightarrow$ dS_3, edge $\rightarrow$ dS_2-JT, sine-dilaton $\rightarrow$ "fake" dS) is flagged, with the gate question then becoming *which placement maps to the 4D cosmological-horizon physics that sets* a_0. The sweep closed by pre-registering the calculation nobody had published.

The discriminator (agent S). That calculation was then run — pre-registered, both placements on identical machinery, all seven banked spectral-weight scripts rerun verbatim first, an independent mpmath pipeline agreeing to 3.6×10^{-17}: the late-time matter two-point function at each placement versus the de Sitter quasinormal ladder. **The center reproduces the dS QNM ladder exactly**: purely damped (oscillation $\sim 10^{-15}$), $\Gamma_n = \sinh((\Delta + n)\lambda)$ to four digits (including an unsupervised matrix-pencil extraction), probe-dimension offset and Δ-free spacing as de Sitter requires, $\geq$19 e-folds deep, $\mathrm{Re}\,\omega = 0$ selecting $\theta = \pi/2$ *uniquely* at finite λ. **The edge fails structurally, under both dimensional matchings** (dS_3 and dS_2-JT — the caveat was run, not waved at): decay locked at $t^{-3/2}$ at coefficient level (Watson's lemma on the square-root spectral edge; Δ- and q-independent; no clock rescues it); one-sided, zero-temperature spectral support where every dS static patch is thermal; QNM rungs exiting the support entirely below $\varepsilon_c \approx \Delta\lambda$ (exact threshold $\cosh(\Delta\lambda) = \sec\varepsilon$). The locked wording, both directions: **CONTESTED-TERMINAL stands at the algebra level** — nothing here derives the placement — and **the contest collapses toward the MOND-favorable center for the sign-relevant observable**, the edge camp's sole rescue (retreat to a triple-scaling limit) severing its own anti-MOND p = 3/5 reading by the same stroke. NOT unlocked: Z, a_0, the unconditional sign. The center's weaknesses are recorded with equal force: its dS-thermal reading runs through the fake-time dictionary; the no-ringing selector degrades to $O(\lambda)$ semiclassically; at finite λ the ladder is an intermediate asymptotic; and this is an internal, unpublished discriminator — it arms the program for the moment the field publishes the same calculation, but cannot by itself flip a literature-level verdict.

7.8 Link 9, and the closure statement

Link 9 — the empirical perimeter (DATA), all owned end to end (Section 5): SPARC 0.105 dex at the framework a_0; the lensing type split at 6.8σ from the program's own 181,477-lens re-measurement; wide binaries degeneracy-limited, the DR4 fork pre-registered three ways and reshaped by Link 6; the solar system as the great filter (it killed F1/Milgrom-99, instantaneous F4 and face-value superfluid dark matter, and selected the exponential tail inside Milgrom-2022's own class); the dated predictions of Section 8.

The closure statement, both ways at full weight. The chain is continuous from Λ to the named effective law — two links carried by exact theorems (Link 2, and Link 5 in the negative), two by data (Links 4 and 9), one by a conditional published chain (Link 4's rival), one by a named template whose covariant home is now built at the EOM level (Link 6, conditional on Link 5's reservoir), one by a proven-necessary, unique-class partner with its scalar, local-vector, and history-keyed carriers all closed constructively (Link 7 — the exposure unexplained by every explored route, the survivor space named) — and **broken in exactly three places, each bounded by a theorem or a contest rather than by ignorance**: the coefficient (two candidates, empirically degenerate, geometry-null), the mechanism (every worldline and field-bath route closed at theorem + data grade; the target reduced to one object with one property — the khronon medium's σ_{req}, since upgraded to a defined calculation with written acceptance conditions), and the quantum gate (terminal at the algebra; edge-wounded at the observables). A complete theory claim would require closing Links 5 and 8 and adjudicating Link 4. Nothing in this edition closes them — and everything in it makes them smaller and sharper than they have ever been.

8 Falsifiable predictions and decision dates

The program's forward value is that its surviving claims are hostage to named measurements with dates. Every row below was registered before the corresponding data exist.

#	Test	Instrument · date	Outcome forks (pre-registered)
1	**Wide-binary boost**	Gaia DR4 (3D velocities), late 2026–2027	Clean null at ∼3% sensitivity: soft MOND shapes (simple, RAR-as-MI: +13–16%) die; the F4 shape (+4% plateau) survives at/below sensitivity. Detection at +10–15%: the F4 shape dies, soft shapes confirmed. The superfluid-class face-value branch (Section 6.5) predicts +26–37% — killed by either a null or a +10–15% detection; only +25–35% selects it. A confirmed full-amplitude boost is also a ΛCDM kill (Newton predicts 0%). DR4 additionally breaks the boost↔contamination degeneracy that makes DR3 null-informative. *Second-edition reshaping (agent M, Section 7.4):* Milgrom-2022's $\theta(0)$-enhanced EFE suppresses the soft-shape boost ×2–4 (soft shapes 4–7% at $\theta(0) \sim e$): a ∼3% null kills soft-shape M22-MI only if $\theta(0) \lesssim 2$; +10–15% kills the $\theta(0)$-enhanced EFE for all shapes; an intermediate **+4–8%** positively selects M22-style enhanced-EFE modified inertia over both AQUAL-EFE modified gravity and undressed F4.
2	**The knee discriminator**	Gaia DR4, same data	Within the hybrid spec, the wide-binary amplitude measures the knee position: knee above $\sim 7\times10^{-13}$ s^{-1} ($mc^2 > 5\times10^{-28}$ eV) makes wide binaries MONDian; below $\sim 8\times10^{-14}$ s^{-1}, Newtonian. **Standing qualification (Section 6.4): the carrier-pays-the-amplitude build this fork was registered for died pre-assembly, so the row no longer discriminates *for* any live build** — it is retained as a constraint on any future amplitude-external realization. A full-amplitude boost pinches the dressing corridor to p ≈ [0.07, 0.1]; a null is consistent with any corridor p — and the carrier fraction ledger (Section 6.1) is pinched hardest exactly at the discriminator mass.
3	**Solar reflex, second channel**	BepiColombo MORE ranging (∼1 cm), 2026–2028.5+	The framework-normalization residual signal spans ∼0.4 cm to ∼1.2 m across the bracket (window-limited over the published 2.5-yr arc); registered as the Door-IVb future trigger. A persistent decade-scale MORE-class campaign would be decisive for any surviving sub-survival-line amplitude.

continued on next page

#	Test	Instrument · date	Outcome forks (pre-registered)
4	**The a_0(z) hostage**	DESI DR3 w(z), ∼2027	The declining branch's quantitative prediction (bump at z ≈ 0.4; 0.737 zero-lag at z = 3) is conditioned on the DESI $w_0 w_a$ dark energy; DR3 re-fits the input. If evolving dark energy evaporates, the distinctive non-monotonic signature collapses toward the constant fallback.
5	**z ≈ 3 kinematics**	JWST/ALMA halo-free rotation curves + BTFR zero-point, ∼2027	The only clean arbiter of a_0(z): rising $\propto \sqrt{\rho_{\text{total}}}$ (≈ 4.6×) is excluded already and would falsify the kernel if measured; constant = 1.0 is the safe fallback; the declining ansatz predicts 0.65–0.74 (zero-lag 0.737), always quoted with its conditionality string (Step-4 coupling + response-gate selection; full band [0.37, 1.0]). The CMB cannot make this discrimination; z ≈ 3 can.
6	**Sub-a_0 atom interferometry**	Existing 10-m-tower instrument class (demonstrated differential accuracy 1.8–3.8×10^{-11} m s^{-2}, arXiv:2005.11624), protocol new	A calibrated sub-a_0 applied force on free-falling atoms (proper acceleration ≈ 0) tests the mechanism-natural proper-acceleration reading: predicted maximal anomaly 1.6×10^{-10} m s^{-2} (hostile normalization — testable at 4.3–9.2σ per campaign now) or 2.8×10^{-11} (framework — 0.7–1.6σ, needs ×3–5 in sensitivity), with a self-calibrating $\sqrt{F}$ response shape. Asymmetry stated: a null adds a wall to an already-walled corner (the solar-reflex kill does not touch this reading — free-falling atoms are the first laboratory objects at x ≈ 0); a detection would outrank everything else here. Corridor caveat: any frequency-dressed hybrid (N5) predicts a null.
7	**The lensing-RAR split**	KiDS/Euclid-class re-measurements; X-ray CGM stacking	Standing falsifier at 8.6–9.2σ against every property-independent law, this framework's included. The named escapes are quantified: the conversion systematic is measured at $\Delta \log C \approx +0.001$ dex (split-strengthening); the hot-gas escape needs $M_{\text{gas}}/M_* \approx$ 1.1 differential against eROSITA's null below $\log M_* =$ 11. A future cool-gas differential measurement at that amplitude, or a demonstrated type-dependent 2-halo/satellite systematic, are the only named outs; conversely a hybrid whose lensing partner produces the split would have to do so quantitatively, type by type.
8	**Cassini-class quadrupole**	Ephemeris updates (tightening: (3±3) → (1.6±1.8)×10^{-27} s^{-2} from 2014 to 2026)	Standing perimeter gate. Any new covariant host must post Q_2 inside [−2.0, +5.2]×10^{-27} s^{-2} with an RAR-compatible function, robustly across g_{ext} — the test Rows 1 and 3 of Section 3 failed. The trend is hostile to gradual-transition theories.

continued on next page

#	Test	Instrument · date	Outcome forks (pre-registered)
9	**Deep-MOND flattening**	Future deep-RAR data: ultra-low-acceleration rotation curves, ultra-faint dwarfs, weak-lensing RAR extension below the current SPARC band	*New (final pass; agent V's no-kernel corollary, Section 7.3).* Any kernel-type response in the surviving mechanism class is analytic in a^2 at a = 0: **no kernel produces the $\sqrt{a/a_0}$ deep-MOND onset, so μ must flatten to constant + $O(a^2)$ below some acceleration $a_\star$** — a framework-distinctive, falsifiable departure from pure MOND at ultra-low accelerations. Bounded both ways: $a_\star$ must sit below x $\approx$ 0.05 (the current band shows no flattening — the legal-mixture family of Section 7.3 died precisely because its flattening intruded into the band), so a measured flattening floor below the band selects the mechanism class, while clean $\sqrt{a/a_0}$ behavior persisting arbitrarily deep falsifies the kernel-type mechanism (not the effective law itself).

Route-discrimination note (second edition; rows 1 and 6 jointly). The two surviving matter-sector routes are separable by *how* they suppress the wide-binary boost: the N5 corridor by frequency, killing all laboratory/atom-interferometer signatures outright (row 6 then predicts a null); Milgrom-2022 by $\theta(0)$-enhanced EFE at p = 0, leaving the row-6 protocol live. Agents U and W (Sections 7.4–7.5) add no new dated rows: the covariant lift transfers the battery at the <1% level, and the slip partner's gates are structural, not dated. The final pass adds row 9 (the flattening corollary) and one undated fingerprint: the causal construction's transient MOND-boost enhancement ($\times$2.32 for $\sim N_{\rm cyc}$ cycles at deep-MOND x, Section 7.4) — now confronted (agent JJ, addendum): the fingerprint's sign is operator-form-dependent (the literal action EOM gives suppression where the implicit-μ reading gives enhancement), and the tidal-dwarf data retire the naive enhancement transplant at $\sim 11\sigma$ while the construction's own EFE-embedded form passes; the decisive three-way test (isolated young rotators, $\geq$0.3 dex splits) is registered as watch entry 13. The addendum adds two computed forks to the watch registry: the flattening's decisive sample (isolated rotators at $g_{\rm obs} = 0.01$–$0.05\,a_0$; floor $\Rightarrow$ a universal downturn at one $g_{\rm obs}$, EFE $\Rightarrow$ none in isolation), and the high-z $a_0(z)$ fork — published JWST-era kinematics sit at $g_{\rm obs} = 5.7$–$24\,a_0$, where the constant and declining branches are Newtonian-degenerate (radius-blocked: the fork lives beyond every last measured point) and the rising rival hides inside the $\times$7 baryonic-mass bracket (mass-blocked); one deep [CII] rotation curve of a REBELS-class disc past $r_{\rm MOND} \approx 6$–10 kpc splits declining/constant/rising by $\times$2.3–2.4 in asymptotic velocity at fixed baryonic mass, decisive if the mass is independently pinned to $\times$2.

9 Methods appendix: the discipline

The program's results are only as good as the protocol that produced them; the protocol is therefore reported as a result.

Pre-registration. Every confrontation with data was preceded by a committed pre-registration naming the outcomes, thresholds, and kill conditions before the computation ran (examples: the wide-binary stages WB-R1 through WB-3b; the lensing battery LR-0; the F4 SPARC shape test thresholds; the DEW quadrupole fork; the N5 SPARC-alive/dead lines). Stop-triggers were honored when they fired (the WB-2 halt).

The convention-robustness rule (the program's first working rule). No deficit or success claim ships until it has been checked against: the regular-MOND defaults ($a_0 = 1.2\times10^{-10}$, $\Upsilon = 0.5$) versus the framework's own (9.36×10^{-11}, $\Upsilon \approx 0.70$); the fit weighting (the unweighted dex-scatter standard is primary, with weighted results shown); the footing (ρ_{DE} versus ρ_{total}); and the a_0(z) branch. Claims that vanish under a reasonable alternative convention are reported as artifacts, in both directions. Worked instances in this paper: the "a_0 is ~20% low" retraction (Section 2.2); the F4-versus-McGaugh SPARC tie (Section 4.2); the DEW rescue cell reported at full weight despite being a sliver (Section 3, Row 3); the F1 kill checked on both footings (Section 4.1).

Hostility inversion. For confrontations where the program's bias would naturally run favorable, the pre-registration inverts the burden: framework-favorable outcomes owe hostile verification at a higher tier than unfavorable ones. The lensing battery was run under this inversion, which is why its hardening verdict (the strongest framework-unfavorable result in the corpus) is also its best-audited one.

The both-ways rule. Every verdict section in the underlying memos carries both directions at full weight — what would have closed the gate and did not, what would have supported the framework and did not. Framework-favorable findings (the N1 gate opening; the N3 sign result; the N5 corridor) are stated with the same prominence as the kills, and vice versa.

The retraction guard. A mechanized guard (`tools/retraction_guard.sh`) blocks any staged document that uses a retracted claim as live (the banned list includes the per-mode CMB $\Delta\chi^2$ regression, the "CMB selects declining" phrasing, and superseded significance figures). This draft passes the guard. Owned and retracted items remain visible in the corpus with their retraction markers, as audit trail: the wide-binary mass-cubic bug; the two Cassini misstatements (a 10× table misread; a bound-versus-central-value confusion); the lensing $5.0\sigma/2.1\sigma$ bracket superseded by measurement; the trilemma's premature "closed perimeter" (re-closed only after the DEW computation); the MUSE "confirms rising" misreading.

Verification standards. Load-bearing algebra is sympy-verified; numerics are cross-checked by independent routes (the N1 commutator: discontinuity formula versus direct boundary values to $\sim 10^{-10}$; the agentE fit emulation: linearized versus true nonlinear LM, independent solvers, conditioning sweeps; the AQUAL solver: three published anchors). Two pipeline bugs in the lensing re-measurement were caught by validation gates *before* any significance was computed. Literature pins are arXiv ids verified in-session; identification errors found in tasking material were corrected and logged.

Appendix A: commit-hash provenance for the major results

Result	Commit
Trilemma completed; missing object named	`b68d9915`
F1/Milgrom-99 ephemeris kill (×54,000)	`3ad8d4cc`
See-saw no-go; F4 selected	`359ee3f8`
F4 SPARC shape test (survives)	`357bbb7f`
F4 wide-binary EFE; DR4 fork inverted	`e908333e`
Swarm: F4 mechanism refuted (λ^2); covariance memo; 40.5σ lensing wall	`9e15560f`
Hostile Υ check (F4 ranking convention-dependent)	`afcabb5c`
DEW quadrupole computed; perimeter re-closes	`742dc889`

continued on next page

Result	**Commit**
Door I-b: non-perturbative closure (all orders)	`d0c6ea8e`
Door IVb: solar reflex kills instantaneous F4	`51d35ccc`
BepiColombo registered as future trigger	`a1462518`
N1 exact commutator (gate opens)	`0db24d0a`
N4 literature map	`7943202f`
N5 keying discriminator (corridor)	`d6624af1`
N3 scale walls (window empty; MOND-signed)	`cdc3daae`
N2 memory Langevin + non-Huygens synthesis (knee window)	`7e91f2bb`
H4 hostile wall-map (knee band; superfluid class)	`9b46ea81`
H1 candidate matrix (the field is empty)	`4979dd0b`
H3 gauntlet (complementary failure; $q_{\rm ph} = -3/7$)	`812a5b2c`
I fraction–amplitude pre-assembly kill	`b1417008`
J mass-bin phase test (staircase refuted $\approx 7.3\sigma$)	`53cada3a`
WB program synthesis (degeneracy-limited)	`f14cf396`
WB methodology note + a_0-insensitivity	`1a42475d`
WB deprojection MC	`284462b8`
WB orbit-integration prior-sensitivity	`258e625c`
LR pre-registration (hostility inverted)	`9518e6e6`
LR battery on released profiles	`d46ade4a`
LR Sérsic cross-check	`0b0aa080`
LR per-class C hardening; lab bridge	`e5bbc19a`
LR independent stack v4 validation	`dfd587cb`
K re-measurement completed (own-data 6.8σ split); clock-coupled exit closed	`96bae2ed`
First edition (integration + hostile pass)	`23413a7a`
Second edition results	
R DSSYK gate sweep (GATE-UNMOVED; discriminator pre-registered)	`025c0494`
L extended-detector coherence closure ($\varepsilon = r_g/2R_H$)	`5e7bc254`
Q theorem: $T_{\rm DL}$ = Tolman-shifted $T_{\rm GH}$; thermodynamic lane closed	`6dc64ba8`
P Verlinde coefficient audit (conditional chain; empirical degeneracy)	`cc9f50b4`
M Milgrom-2022 gauntlet (template adopted; DR4 fork reshaped)	`bd2207d4`
Derivation chain consolidated (`DERIVATION_CHAIN.md`)	`2250de85`
T geometry null (Z stays data-selected); dS Killing-failure theorem	`e677a340`
U khronon-M22 covariant lift (buildable, unbuilt)	`d91da00a`

continued on next page

Result	Commit
W double-counting theorem; Ψ-slip unique class	`5d055f3d`
S edge-QNM discriminator (EDGE-WOUNDED)	`392b6097`
Final-pass results	
Z type-split second variable: TYPE-IRREDUCIBLE (+0.194 dex at fixed M_*/z/density)	`e8054a5e`
Unified action assembly frame (scaffold + interface conditions)	`c48e4335`
Unified action field equations (δg/δu/δx; the Bianchi spine)	`7f198c8b`
X SK gate: causal conserving EOM built; Theorem X2 (passivity)	`c7f44f40`
V kernel inversion (KERNEL-EXISTS-BUT-ILLEGAL; flattening corollary)	`8a1411d9`
Chain Links 5+6 rewritten to final form	`72646111`
AA sign reconciliation (DEFICIT $\iff$ $M^2 < 2H^2$; V's flag retired)	`c2fc46c0`
BB legal-mixture ESCAPE-CLOSED (boundary theorem to data grade)	`d105df87`
Y scalar slip carrier OBSTRUCTED (four walls; class narrowed)	`8b07b3a4`
LaTeX/Zenodo edition (.tex + 34-pp PDF; 10 passages verified verbatim)	`3770c9c6`
CC: $a_\star$ first confrontation (no flattening; $a_\star \leq 0.107\, a_0$; decisive test registered)	`40116556`
GG: JWST high-z REGIME-INSUFFICIENT; the ×2.3–2.4 branch fork registered	`70b56478`
FF: hostile audit of X (X2 strengthened to nonlinear; signed ledger ACTIVE-secular; three framings corrected)	`93524ad1`
EE: $\sigma_{\rm req}$ scoping — STRUCTURALLY-CAPABLE (pumped khronon); Link 5 = a defined calculation; the b-family theorem	`9def577c`
DD: vector carrier SAME-WALLS; THE KEYING THEOREM; the dial two-carrier-robust	`0c3ff2ee`
II: CMB/E_G audit — KILL of the naive linear-scale slip extension; the second-discriminant requirement	`641021ad`
JJ: the transient fingerprint confronted; the (X-4) operator-form split; TDG transplant dead $\sim 11\sigma$	`25d90036`
KK: nonlocal/history class OBSTRUCTED (static equivalence); the convergence line retired	`480f8833`

References

Published works (arXiv ids as pinned in the program's memos; pre-arXiv classics by journal):

- Milgrom, M. 1983, ApJ 270, 365. — 1994, Ann. Phys. 229, 384, arXiv:astro-ph/9303012. — 1999, Phys. Lett. A 253, 273, arXiv:astro-ph/9805346. — 2011, Acta Phys. Pol. B 42, 2175, arXiv:1111.1611. — 2017, arXiv:1703.06110. — 2022, PRD 106, 064060, arXiv:2208.07073. — 2023, arXiv:2310.14334.
- Bekenstein, J. 2008, arXiv:0809.2790.

- Skordis, C. & Złośnik, T. 2021, PRL 127, 161302 (AeST).
- Babichev, E., Deffayet, C. & Esposito-Farèse, G. 2011, arXiv:1106.2538.
- Bruneton, J.-P. & Esposito-Farèse, G. 2007, arXiv:gr-qc/0607055; restatement arXiv:2503.11174.
- Deffayet, C., Esposito-Farèse, G. & Woodard, R. 2011, PRD 84, 124054, arXiv:1106.4984; Deffayet, C. & Woodard, R. 2026, JCAP 04 (2026) 081, arXiv:2512.10513; Woodard, R. 2014, arXiv:1403.6763; Soussa, M. & Woodard, R. 2003, arXiv:astro-ph/0302030, astro-ph/0307358.
- Desmond, H., Hees, A. & Famaey, B. 2024, MNRAS 530, 1781, arXiv:2401.04796. Park, R., Hees, A., Famaey, B. et al. 2026, arXiv:2602.17884. Hees, A. et al. 2014, arXiv:1402.6950; 2016, MNRAS 455, 449, arXiv:1510.01369; 2016, AJ 152, 94, arXiv:1604.03180.
- Fienga, A. et al. 2009, arXiv:0906.3962. Pioneer review: arXiv:1001.3686. Bepi-Colombo/MORE: arXiv:2111.04499, 2211.04881. Durante, D. et al. 2020, GRL 47, e2019GL086572.
- Deser, S. & Levin, O. 1997, CQG 14, L163, arXiv:gr-qc/9706018. Bunch, T. & Davies, P. 1978, Proc. R. Soc. A 360, 117. Spradlin, M., Strominger, A. & Volovich, A. 2001, arXiv:hep-th/0110007. Bros, J. & Moschella, U. 1996, arXiv:gr-qc/9511019. Allen, B. 1985, PRD 32, 3136; Allen, B. & Folacci, A. 1987, PRD 35, 3771. López-Ortega, A. 2006, arXiv:gr-qc/0605027. Takagi, S. 1986, Prog. Theor. Phys. Suppl. 88, 1. Letaw, J. 1981, PRD 23, 1709. Obadia, N. & Milgrom, M. 2007, arXiv:gr-qc/0701130.
- Lin, S.-Y. & Hu, B.-L. 2006, arXiv:gr-qc/0507054; 2007, arXiv:gr-qc/0611062. Hu, B.-L., Paz, J. & Zhang, Y. 1992, PRD 45, 2843. Ford, G., Lewis, J. & O'Connell, R. 1988, PRA 37, 4419. Kaplanek, G. & Burgess, C. 2020, arXiv:1912.12951, 1912.12955.
- Mino, Y., Sasaki, M. & Tanaka, T. 1997, arXiv:gr-qc/9606018. Quinn, T. & Wald, R. 1997, arXiv:gr-qc/9610053. Quinn, T. 2000, arXiv:gr-qc/0005030. Gralla, S. & Wald, R. 2008, arXiv:0806.3293. Poisson, E., Pound, A. & Vega, I. 2011, arXiv:1102.0529. Burko, L., Harte, A. & Poisson, E. 2002, PRD 65, 124006, arXiv:gr-qc/0201020. Haas, R. & Poisson, E. 2005, arXiv:gr-qc/0411108. Galley, C. & Hu, B.-L. 2009, arXiv:0801.0900. Galley, C., Hu, B.-L. & Lin, S.-Y. 2005, arXiv:gr-qc/0505085. Galley, C. 2013, arXiv:1210.2745. Chu, Y.-Z. 2016, arXiv:1611.00018. Chu, Y.-Z. & Starkman, G. 2011, arXiv:1108.1825. Starobinsky, A. & Yokoyama, J. 1994, arXiv:astro-ph/9407016.
- Gundlach, J. et al. 2007, PRL 98, 150801. Abramovici, A. & Vager, Z. 1986, PRD 34, 3240. Ignatiev, A. 2007, arXiv:gr-qc/0612159. Peters, A., Chung, K. & Chu, S. 1999, Nature 400, 849. Asenbaum, P. et al. 2020, PRL 125, 191101, arXiv:2005.11624. Touboul, P. et al. (MICROSCOPE) 2022, PRL 129, 121102, arXiv:2209.15487, 2209.15488. Will, C. 2014, arXiv:1403.7377. Damour, T. & Donoghue, J. 2010, arXiv:1007.2792. Kapner, D. et al. 2007, arXiv:hep-ph/0611184. Wagner, T. et al. 2012, arXiv:1207.2442. Cohen, A., Kaplan, D. & Nelson, A. 1999, arXiv:hep-th/9803132. Blinov, N. & Draper, P. 2021, arXiv:2107.03530.
- Chae, K.-H. 2023, ApJ 952, 128, arXiv:2305.04613. Banik, I. et al. 2024, MNRAS 527, 4573, arXiv:2311.03436. El-Badry, K., Rix, H.-W. & Heintz, T. 2021, MNRAS 506, 2269.
- Brouwer, M. et al. 2021, A&A 650, A113, arXiv:2106.11677. Zhang, Y. et al. 2025 (eROSITA eRASS CGM), arXiv:2411.19945. Mistele, T. et al. 2023, arXiv:2303.08560;

2024, ApJL 969, L3, arXiv:2406.09685. Moster, B., Naab, T. & White, S. 2013, arXiv:1205.5807. Behroozi, P., Wechsler, R. & Conroy, C. 2013, arXiv:1207.6105. Duffy, A. et al. 2008, arXiv:0804.2486. Dutton, A. & Macciò, A. 2014, arXiv:1402.7073. Mandelbaum, R. et al. 2016, arXiv:1509.06762.

- Berezhiani, L. & Khoury, J. 2015, PRD 92, 103510, arXiv:1507.01019 (and PLB 753, 639, arXiv:1506.07877); Khoury, J. 2016, PRD 93, 103533, arXiv:1602.05961; Berezhiani, L. et al. 2018, arXiv:1711.05748; Sharma, A., Khoury, J. & Lubensky, T. 2019, JCAP 05, 054, arXiv:1809.08286; Berezhiani, L., Cintia, G. & Khoury, J. 2023, PRD 107, 123010, arXiv:2212.10577; review 2026, Phys. Rep., arXiv:2505.23900. Lisanti, M. et al. 2019, PRD 100, 083009, arXiv:1812.08169; arXiv:1911.12365; arXiv:2003.07324. Mistele, T., McGaugh, S. & Hossenfelder, S. 2022, A&A 664, A40, arXiv:2201.07282. Strong lensing: arXiv:1809.00840. Cherenkov: arXiv:2103.16954, 2208.14308. Hyperbolicity: Hertzberg, M., Litterer, J. & Shah, N. 2021, JCAP 11, 015, arXiv:2105.02241. Structure: arXiv:2009.03003, 2001.02591, 2109.01011. Volovik, G. 2025, arXiv:2410.04392. Ferreira, E. et al. 2018, arXiv:1810.09474.
- Other matrix rows (§6.3): Famaey, B., Khoury, J. & Penco, R. 2018, JCAP 03, 038, arXiv:1712.01316 (+ arXiv:1912.07626). Lee, J.-W. et al. 2019, arXiv:1901.00305. Kamada, A. 2026, arXiv:2605.20217. Blanchet, L. & Le Tiec, A. 2008–09, arXiv:0804.3518, 0901.3114; bimetric arXiv:1504.00870; completion failure arXiv:2302.02690. Hees, A., Famaey, B. & Bertone, G. 2017, PRD 95, 064019, arXiv:1702.04358. Boran, S. et al. 2018, PRD 97, 041501, arXiv:1710.06168.
- ULDM bounds: Rogers, K. & Peiris, H. 2021, PRL 126, 071302, arXiv:2007.12705. Dalal, N. & Kravtsov, A. 2022, arXiv:2203.05750. Nadler, E. et al. 2021, PRL 126, 091101, arXiv:2008.00022. Hložek, R. et al., arXiv:1410.2896, 1708.05681. Laguë, A. et al., arXiv:2104.07802. eROSITA clusters 2025, arXiv:2502.03353. Rogers, K. et al. 2023, arXiv:2301.08361. Kobayashi, T. et al. 2017, arXiv:1708.00015. Smarra, C. et al. (EPTA) 2023, PRL 131, 171001, arXiv:2306.16228; PPTA update arXiv:2605.02172. Stott, M. 2020, arXiv:2009.07206. Davoudiasl, H. & Denton, P. 2019, arXiv:1904.09242. Marsh, D. & Niemeyer, J. 2019, arXiv:1810.08543.
- a_0(z) front: Ciocan, B., Bouché, N. et al. 2026 (MUSE-DARK III), arXiv:2604.22613. Mayer, A. et al. 2023, MNRAS 518, 257, arXiv:2206.04333. Del Popolo, A. & Chan, M. 2024, arXiv:2405.01841. Tiley, A. et al. 2019, arXiv:1810.07202. Tian, Y. et al. 2024, arXiv:2402.12016. Li, P. et al. 2024, arXiv:2411.09735.
- Adjacents: McCulloch, M. 2007, arXiv:astro-ph/0612599 (and the skeptical analysis Renda, M. 2019, arXiv:1908.01589). Smolin, L. 2017, arXiv:1704.00780. Verlinde, E. 2017, arXiv:1611.02269.
- Final-pass additions (§7). *Causal nonconservative mechanics:* Galley, C. 2013, PRL 110, 174301, arXiv:1210.2745. Galley, C., Tsang, D. & Leibovich, A. 2014, arXiv:1412.3082. *DHOST classification and constraints:* Langlois, D. & Noui, K. 2016, arXiv:1510.06930. Langlois, D. 2019, arXiv:1811.06271. Kobayashi, T. 2019, arXiv:1906.03020. Ezquiaga, J. M. & Zumalacárregui, M. 2017, arXiv:1710.05901. Langlois, D., Saito, R., Yamauchi, D. & Noui, K. 2018, arXiv:1711.07403. Creminelli, P. et al. 2018, arXiv:1809.03484; 2019, arXiv:1910.14035. Crisostomi, M. & Koyama, K. 2018, arXiv:1711.06661. Dima, A. & Vernizzi, F. 2018, arXiv:1712.04731. *dS Källén–Lehmann positivity:* Hogervorst, M., Penedones, J. & Vaziri, K. 2021, arXiv:2107.13871. Loparco, M. et al. 2023, arXiv:2306.00090.
- Second-edition additions (§7). *Horizon thermodynamics:* Jacobson, T. 1995, PRL 75, 1260, arXiv:gr-qc/9504004. Guedens, R., Jacobson, T. & Sarkar, S. 2011, arXiv:1112.6215.

Narnhofer, H., Peter, I. & Thirring, W. 1996, IJMP B10, 1507. *Coefficient contest:* Dai, D.-C. & Stojkovic, D. 2017, JHEP 11, 007, arXiv:1710.00946. Yoon, Y. 2020, arXiv:2003.03198; 2024, Phys. Dark Universe 45, 101551. Yoon, Y., Park, J.-C. & Hwang, H. 2022, arXiv:2206.11685. Diez-Tejedor, A. et al. 2018, MNRAS 477, 1285, arXiv:1612.06282. *Extended-detector closure:* McGaugh, S., Lelli, F. & Schombert, J. 2016, PRL 117, 201101, arXiv:1609.05917. Lelli, F. et al. 2017, arXiv:1610.08981. Williams, J., Turyshev, S. & Boggs, D. 2012, CQG 29, 184004, arXiv:1203.2150. *Covariant home:* Gümrükçüoğlu, A. E., Saravani, M. & Sotiriou, T. 2018, arXiv:1711.08845. Oost, J., Mukohyama, S. & Wang, A. 2018, arXiv:1802.04303. Gupta, A. et al. 2021, arXiv:2104.04596. Blanchet, L. & Marsat, S. 2011, PRD 84, 044056, arXiv:1107.5264. Flanagan, É. 2023, arXiv:2302.14846. Bonetti, M. & Barausse, E. 2015, PRD 91, 084053, arXiv:1502.05554. Belgacem, E., Dirian, Y., Foffa, S. & Maggiore, M. 2017, arXiv:1712.07066. Famaey, B. & Durakovic, A. 2025, arXiv:2501.17006. *Partner uniqueness:* GW170817 speed bound, arXiv:1710.05834. Saltas, I., Sawicki, I., Amendola, L. & Kunz, M. 2014, PRL 113, 191101, arXiv:1406.7139; 2016, arXiv:1612.02002. Battye, R. & Moss, A. 2007, arXiv:astro-ph/0703744; Battye, R. & Pearson, J. 2013, arXiv:1301.5042. Mistele, T. 2024, arXiv:2408.02725. *Quantum gate:* Narovlansky, V. & Verlinde, H. 2023, arXiv:2310.16994. Narovlansky, V. 2025, arXiv:2506.02109. Okuyama, K. 2025, arXiv:2505.08116. Marini, A., Qi, X.-L. & Verlinde, H. 2026, arXiv:2604.21014. Goto, K., Milekhin, A., Verlinde, H. & Xu, W. 2026, arXiv:2605.03037. Lin, H. & Susskind, L. 2022, arXiv:2206.01083. Jafferis, D. et al. 2013, arXiv:1305.5523. Maldacena, J., Turiaci, G. & Yang, Z. 2019, arXiv:1904.01911.

Repository memos underlying each section (relative to `real_research/`): §2 — `CONVENTION_LOCK.md`, `FRAMEWORK_A0_LAW_OF_NATURE_2026-06-06.md`, `A0Z_KERNEL_STANDING_2026-06-06.md`; §3 — `CASSINI_QUADRUPOLE_CONSTRAINT.md`, `REALIZATION_REDTEAM_galileon_singular_surface_2026-06-06.md`, `reviews/toe_law/TOE_TRILEMMA.md`, `reviews/toe_law/agentC_covariance_memo.md`, `reviews/toe_law/agentD_dew_quadrupole.md`; §4 — `reviews/toe_law/MI_BATH_TAIL_CONSTRAINT.md`, `MI_COUPLING_FAMILY.md`, `TOE_STATUS_AND_DOORS.md`, `agentA/B/E/F/G` and `agentN1-N5` memos; §5 — `reviews/public_data/WB_PROGRAM_SUMMARY.md`, `WB_METHODOLOGY_NOTE_DRAFT.md`, `reviews/lensing_rar/lr_battery_results.md`, `agentH_perclass_C.md`, `lr_stack_v4.out`, `reviews/toe_law/f4_lensing_wall.out`; §6 — `reviews/toe_law/NONHUYGENS_DOOR_SYNTHESIS.md`, `agentH4_hostile_walls.md`, `agentH1_candidate_matrix.md`, `agentI_fraction_amplitude.md` (+`.out`), `agentH3_gauntlet.md` (+`agentH3_gauntlet.out`, `agentH3_typesplit.out`), `agentJ_massbin_phase.md`; §7 — `reviews/toe_law/DERIVATION_CHAIN.md`, `agentQ_jacobson_DL.md`, `agentP_verlinde_coefficient.md`, `agentT_zgeometry.md`, `agentL_extended_coherence.md`, `agentM_milgrom2022_gauntlet.md`, `agentU_khronon_m22.md`, `agentW_partner_uniqueness.md`, `agentR_dssyk_gate_2026.md`, `agentS_edge_qnm.md`, and (final pass) `agentX_sk_gate.md`, `agentV_kernel_inversion.md`, `agentAA_sign_reconciliation.md`, `agentBB_legal_mixture.md`, `agentY_psislip_construction.md`, `UNIFIED_ACTION_ASSEMBLY.md`, `reviews/lensing_rar/agentZ_second_variable.md` (each with its `.py`/`.out` artifacts); §8 — `FALSIFICATION_MATRIX.md` and the watch registry.

This whitepaper presents a falsification map, exact no-go results, and a specification. It does not present a theory of everything, a derivation of $Z = 5.789$, or a mechanism for the kernel; per the program's own boundary, anyone claiming those from the current state would be manufacturing. The second edition adds a derivation chain and a blueprint — named in both halves, not found, not built. The final pass assembles the blueprint into a candidate-theory scaffold — named, specified, gated, not validated — with the matter half built at the equation-of-motion level conditional on Link 5's reservoir; the boundary stands: Z underived, $\sigma_{\rm req}$ underived, the quantum gate unmoved at the algebra.